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Department of Computer Engineering

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UID no.	2023300222	2024301013
Experiment No.	6 - Mini Project	

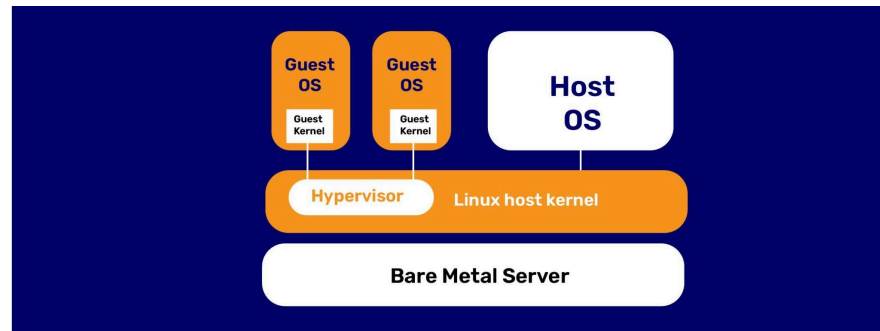
AIM:	To design and develop a scalable full-stack web application (CodeThrone) and implement a complete DevOps-based CI/CD pipeline using virtualization, containerization, version control, continuous integration, code quality analysis, and infrastructure management tools such as KVM, Docker, Jenkins, SonarQube, Multipass, and LXD, ensuring automated build, testing, deployment, and monitoring of the application.
Mini Project	
PROBLEM STATEMENT :	Modern web applications require scalable architecture, automated deployment, efficient resource utilization, and continuous quality assurance. Traditional deployment methods are manual, error-prone, and lack scalability and automation. The objective of this mini project is to develop a full-stack web application named CodeThrone and integrate DevOps practices using multiple technologies to achieve: ● Automated CI/CD pipeline ● Scalable virtualized and containerized environments ● Efficient version control and collaboration ● Continuous code quality analysis ● Reliable deployment and monitoring The system should demonstrate end-to-end automation from code commit to deployment while ensuring maintainability, scalability, and performance.
THEORY:	1. Virtualization using KVM Virtualization is a technology that allows multiple operating systems to run on a single physical machine by abstracting hardware resources through a hypervisor. Kernel-based Virtual Machine (KVM) is a type-1 hypervisor integrated into



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the Linux kernel. It converts the Linux system into a hypervisor, enabling the creation and management of virtual machines (VMs). Each VM runs its own operating system and behaves like a physical machine.

In this project, virtualization was used to create an isolated development environment where all DevOps tools such as Docker, Jenkins, and SonarQube were deployed. This ensured system-level isolation, reproducibility, and independence from the host machine.



Key concepts:

- Hypervisor: Manages VMs and hardware resources
- Guest OS: Operating system inside VM
- Host OS: Main system running virtualization
- Resource isolation: CPU, memory, and storage allocation

Advantages:

- Strong isolation
- Hardware abstraction
- Easy testing and deployment

In this project, virtualization forms the foundation layer where all DevOps activities are performed.

2. Containerization using Docker

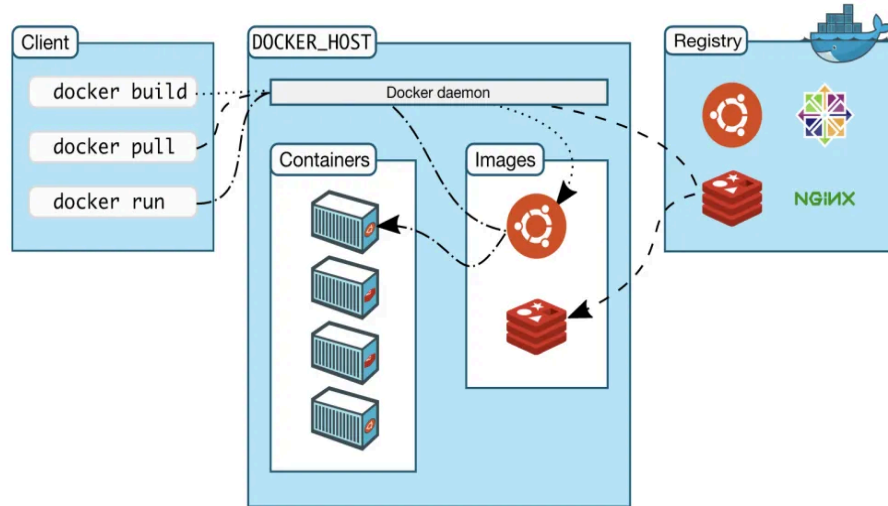
Containerization is a lightweight virtualization technique that packages applications and their dependencies into containers.

Docker is a platform that enables building, shipping, and running



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applications inside containers. Unlike virtual machines, containers share the host OS kernel, making them faster and more efficient.



In this project:

- CodeThrone application was containerized using Docker
- Dockerfile was created to define application build steps
- Containers were used to run frontend, backend, Jenkins, and SonarQube

Key components:

- Docker Image: Blueprint of application
- Docker Container: Running instance of image
- Dockerfile: Instructions to build image
- Docker Engine: Runtime for containers

Docker benefits:

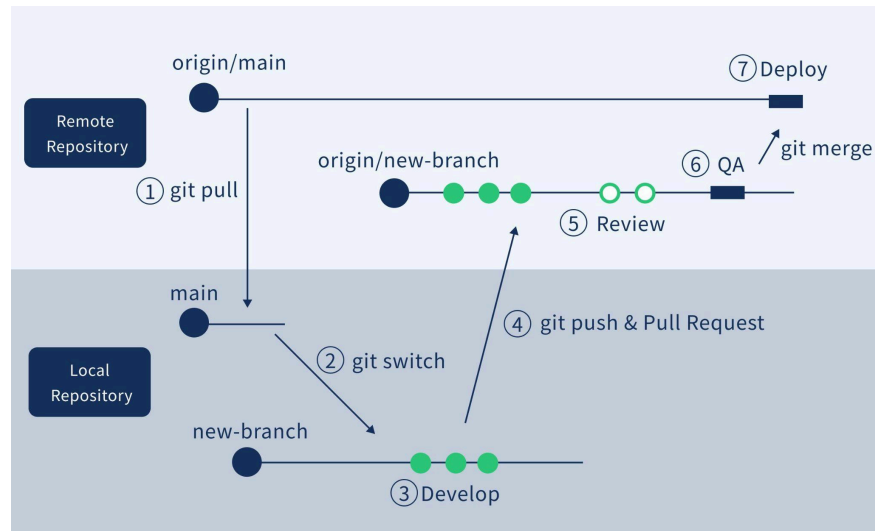
- Portability (runs anywhere)
- Fast deployment
- Efficient resource utilization
- Microservices compatibility

Docker enabled consistent deployment of the CodeThrone application across environments.



3. Version Control using Git

Version control systems track changes in source code and enable collaborative development.



Git is a distributed version control system, and GitHub is used as a remote repository platform.

In this project:

- CodeThrone source code is maintained on GitHub
- Code changes are tracked through commits
- Branching is used for development and feature management
- Jenkins integrates with GitHub for automated builds

Key operations:

- git clone → copy repository
- git commit → save changes
- git push → upload changes
- git pull → update local code

Advantages:

- Collaboration among developers
- Code history tracking

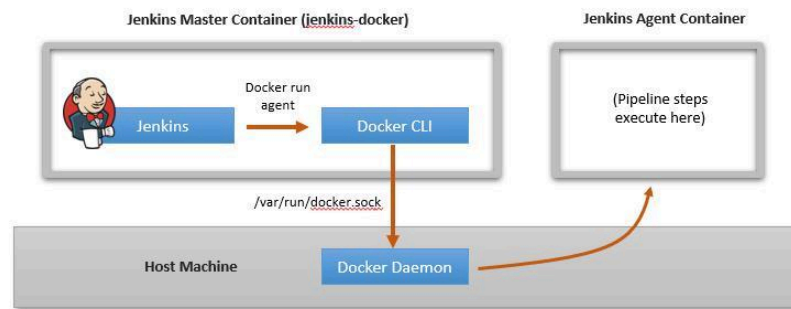


- Easy rollback
- Integration with CI/CD tools

Git acts as the starting point of the CI/CD pipeline.

4. Continuous Integration & Deployment using Jenkins

Continuous Integration (CI) is the practice of automatically building and testing code whenever changes are made. Continuous Deployment (CD) automates application deployment.



Jenkins is an open-source automation server used to implement CI/CD pipelines.

In this project:

- Jenkins was deployed using Docker
- Jenkins pipeline was configured using Jenkinsfile
- GitHub repository was integrated with Jenkins
- Automated steps include:
 - Code checkout
 - Dependency installation
 - Build process
 - Application execution

Pipeline stages:

1. Checkout code from GitHub
2. Install dependencies (npm install)
3. Build application



4. Run application

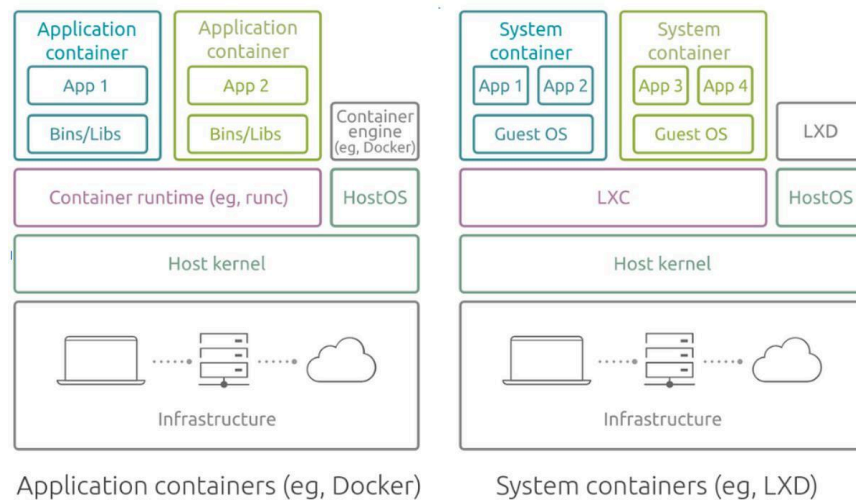
Benefits:

- Automation of repetitive tasks
- Faster development cycles
- Reduced human errors
- Continuous feedback

Jenkins forms the core automation engine of the project.

5. Multipass and LXD (Virtual Machine & Container Management)

Multipass is a lightweight VM manager used to quickly create Ubuntu virtual machines. LXD is a system container manager that provides VM-like containers.



In this project:

- Multipass was used to create VM instances for development
- LXD was used to create container-based environments
- Docker was installed inside LXD containers

Key features:

- Multipass:
 - Quick VM provisioning



- Cloud-init support
- LXD:
 - Lightweight containers
 - Snapshots and profiles
 - Network configuration

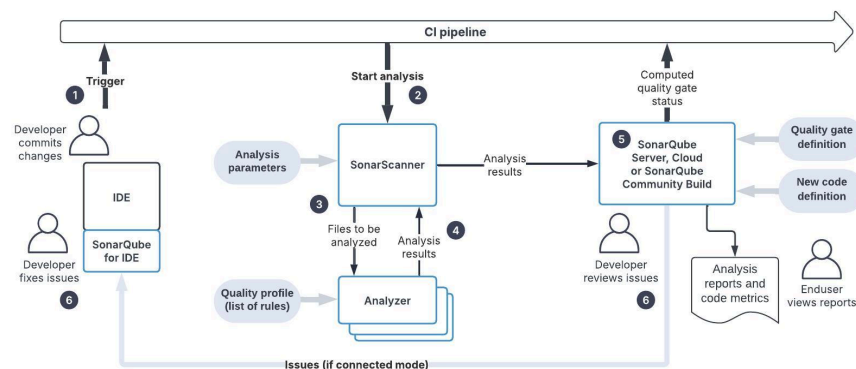
Advantages:

- Hybrid virtualization (VM + container)
- Efficient resource utilization
- Easy environment management

This layer enhances flexibility in infrastructure management.

6. Code Quality Analysis using SonarQube

SonarQube is a static code analysis tool used to inspect code quality and detect bugs, vulnerabilities, and code smells.



In this project:

- SonarQube was deployed using Docker
- CodeThrone project was analyzed using Sonar Scanner
- Quality metrics such as:
 - Bugs
 - Vulnerabilities
 - Code smells
 - Duplication



were generated

Key concepts:

- Static analysis (without execution)
- Quality gates
- Code metrics

Benefits:

- Improved code quality
- Early bug detection
- Security enhancement
- Maintainability tracking

SonarQube ensures the reliability and quality of the application.

7. Integration of All Technologies (DevOps Pipeline)

The project integrates all technologies into a complete DevOps workflow:

1. Developer writes code → GitHub
2. Jenkins pulls code → CI pipeline triggered
3. Docker builds application → container deployment
4. SonarQube analyzes code quality
5. Application runs inside VM (Multipass / LXD)

This demonstrates a complete CI/CD lifecycle.

8. About CodeThrone Application

CodeThrone is a full-stack competitive programming platform that allows users to practice coding, participate in contests, and collaborate via discussions and chat .

The system follows a microservices-based architecture with:

- Frontend: React + TypeScript
- Backend: Node.js + Express
- Database: MongoDB
- Real-time communication: Socket.io



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	The platform includes: ● Coding contests and leaderboards ● Code editor and problem-solving environment ● Chat and discussion modules ● User profiles and analytics The architecture is scalable, modular, and designed for high performance and real-time interaction.
PROCEDURE:	The implementation of the CodeThrone project and its DevOps pipeline was carried out in a structured and phased manner. Each technology was configured step-by-step and integrated into the overall system. Phase 1: Virtualization Setup using KVM 1. Installed virtualization tools on host system: <pre>sudo apt update sudo apt install qemu-kvm libvirt-daemon-system libvirt-clients bridge-utils virt-manager</pre> 2. Verified virtualization support: <pre>kvm-ok</pre> 3. Created and configured a Virtual Machine using Virtual Machine Manager. 4. Installed Ubuntu OS inside VM. 5. Configured network settings and verified internet connectivity. This VM acts as the primary DevOps environment for all further experiments. Phase 2: Containerization using Docker 1. Installed Docker inside VM: <pre>sudo apt update sudo apt install docker.io -y</pre> 2. Started and enabled Docker service: <pre>sudo systemctl start docker sudo systemctl enable docker</pre> 3. Verified installation: <pre>docker --version</pre>



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4. Cloned CodeThrone project:

```
git clone  
https://github.com/CodingEnthusiastic/LeetCode-Throne.git  
cd LeetCode-Throne
```

5. Created Dockerfile for application.

6. Built Docker image:

```
docker build -t blitz-app .
```

7. Ran container:

```
docker run -d -p 3000:3000 --name blitz-container  
blitz-app
```

8. Verified application:

```
curl localhost:3000
```

Application successfully containerized and deployed.

Phase 3: Version Control using Git

1. Configured Git:

```
git config --global user.name "Your Name"  
git config --global user.email "your@email.com"
```

2. Cloned repository:

```
git clone  
https://github.com/CodingEnthusiastic/LeetCode-Throne.git
```

3. Performed operations:

```
git add .  
git commit -m "Updated Jenkinsfile"  
git push
```

GitHub repository acts as the source trigger for CI/CD pipeline.

Phase 4: CI/CD Pipeline using Jenkins

1. Pulled Jenkins Docker image:

```
docker pull jenkins/jenkins:lts
```

2. Ran Jenkins container:

```
docker run -d -p 8080:8080 -p 50000:50000 \  
-v jenkins_home:/var/jenkins_home \
```



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```
-v /var/run/docker.sock:/var/run/docker.sock \  
--name jenkins jenkins/jenkins:lts
```

3. Accessed Jenkins:
`http://<VM-IP>:8080`
4. Installed required plugins:
 - Git Plugin
 - Pipeline Plugin
5. Created Pipeline Job:
 - Connected to GitHub repository
 - Added Jenkinsfile
6. Jenkinsfile stages implemented:
 - Checkout
 - Install Dependencies
 - Build Application
 - Run Application
7. Executed pipeline and verified logs.

CI/CD pipeline successfully automated build and deployment.

Phase 5: Multipass & LXD Setup

Multipass

1. Installed Multipass:
`sudo snap install multipass`
2. Created VM:
`multipass launch --name devops-instance --disk 20G --mem 4G --cpus 2`
3. Accessed VM:
`multipass shell devops-instance`

LXD

1. Installed LXD:
`sudo apt install lxd -y`
2. Initialized LXD:
`sudo lxd init`
3. Created container:



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```
lxc launch ubuntu:22.04 dev-container
```

4. Accessed container:

```
lxc exec dev-container -- bash
```

5. Installed Docker inside LXD container.

Demonstrated hybrid infrastructure (VM + container).

Phase 6: SonarQube Setup (Code Quality Analysis)

1. Pulled SonarQube image:

```
docker pull sonarqube:lts-community
```

2. Ran container:

```
docker run -d -p 9000:9000 --name sonarqube  
sonarqube:lts-community
```

3. Accessed SonarQube:

```
http://<VM-IP>:9000
```

4. Created project and generated token.

5. Installed Sonar Scanner.

6. Ran analysis:

```
sonar-scanner
```

7. Viewed results on dashboard.

Code quality metrics successfully analyzed.



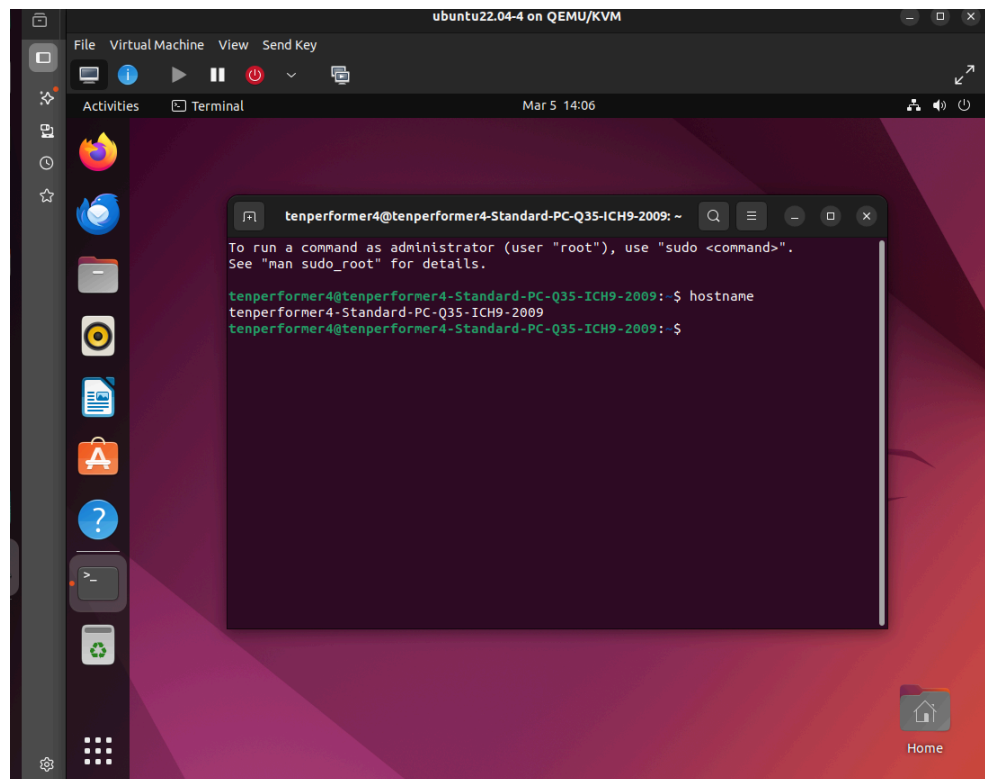
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SCREENSHOTS:

Virtualization (KVM)

```
mca@mca-ThinkCentre-neo-50s-Gen-3:~$ sudo apt update
[sudo] password for mca:
Hit:1 https://download.docker.com/linux/ubuntu jammy InRelease
Hit:2 https://dl.google.com/linux/chrome/deb stable InRelease
Hit:3 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Hit:5 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:6 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
4 packages can be upgraded. Run 'apt list --upgradable' to see them.
mca@mca-ThinkCentre-neo-50s-Gen-3:~$ sudo apt install qemu-kvm libvirt-daemon-system libvirt-clients bridge-utils virt-manager
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'qemu-system-x86' instead of 'qemu-kvm'
bridge-utils is already the newest version (1.7-1ubuntu3).
virt-manager is already the newest version (1:4.0.0-1).
libvirt-clients is already the newest version (8.0.0-1ubuntu7.15).
libvirt-daemon-system is already the newest version (8.0.0-1ubuntu7.15).
qemu-system-x86 is already the newest version (1:6.2+dfsg-2ubuntu6.28).
The following packages were automatically installed and are no longer required:
  libwpe-1.0-1 libwpebackend-fdo-1.0-1 ubuntu-fan
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 4 not upgraded.
mca@mca-ThinkCentre-neo-50s-Gen-3:~$ sudo systemctl status libvirtd
● libvirtd.service - Virtualization daemon
   Loaded: loaded (/lib/systemd/system/libvirtd.service; enabled; vendor preset: enabled)
   Active: active (running) since Thu 2026-03-05 13:17:20 IST; 35min ago
     TriggeredBy: ● libvirtd-admin.socket
                  ● libvirtd-ro.socket
                  ● libvirtd.socket
    Docs: man:libvirtd(8)
          https://libvirt.org
   Main PID: 874 (libvirtd)
     Tasks: 21 (limit: 32768)
    Memory: 30.7M
      CPU: 510ms
   CGroup: /system.slice/libvirtd.service
           └─ 874 /usr/sbin/libvirtd
             1352 /usr/sbin/dnsmasq --conf-file=/var/lib/libvirt/dnsmasq/default
             1354 /usr/sbin/dnsmasq --conf-file=/var/lib/libvirt/dnsmasq/default

Mar 05 13:17:20 mca-ThinkCentre-neo-50s-Gen-3 systemd[1]: Started Virtualization daemon: libvirtd.
Mar 05 13:17:21 mca-ThinkCentre-neo-50s-Gen-3 dnsmasq[1352]: started, version 2.9.4
```





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```
Activities Virtual Machine Manager Mar 5 14:13
ubuntu22.04.4 on QEMU/KVM
File Virtual Machine View Send Key
Activities Terminal Mar 5 14:13

tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009: ~
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ hostname
tenperformer4-Standard-PC-Q35-ICH9-2009
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ sudo snap install multi
pass
[sudo] password for tenperformer4:
Sorry, try again.
[sudo] password for tenperformer4:
multipass 1.16.1 from Canonical** installed
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ multipass version
multipass 1.16.1
multipassd 1.16.1
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$
```

Docker

```
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009: ~
Scanning linux images...
Running kernel seems to be up-to-date.
Restarting services...
systemctl restart udisks2.service
No containers need to be restarted.
No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@devops-instance:~$ git --version
git version 2.43.0
ubuntu@devops-instance:~$ git clone https://github.com/CodingEnthusiastic/LeetCode-Throne.git
Cloning into 'LeetCode-Throne'...
remote: Enumerating objects: 2035, done.
remote: Counting objects: 100% (355/355), done.
remote: Compressing objects: 100% (167/167), done.
remote: Total 2035 (delta 267), reused 228 (delta 184), pack-reused 1680 (from 2)
Receiving objects: 100% (2035/2035), 43.45 MiB | 17.45 MiB/s, done.
Resolving deltas: 100% (1415/1415), done.
ubuntu@devops-instance:~$ cd LeetCode-Throne
ubuntu@devops-instance:~/LeetCode-Throne$ sudo apt install docker.io -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
docker.io is already the newest version (28.2.2-0ubuntu1-24.04.1).
0 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.
ubuntu@devops-instance:~/LeetCode-Throne$ sudo systemctl start docker
sudo systemctl enable docker
ubuntu@devops-instance:~/LeetCode-Throne$ docker --version
Docker version 28.2.2, build 28.2.2-0ubuntu1-24.04.1
ubuntu@devops-instance:~/LeetCode-Throne$ sudo usermod -aG docker ubuntu
ubuntu@devops-instance:~/LeetCode-Throne$ exit
multipass shell devops-instance
logout
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$
```



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```
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009: ~  
GNU nano 6.2 Dockerfile *  
FROM node:18  
  
WORKDIR /app  
  
COPY . .  
  
RUN npm install  
  
EXPOSE 3000  
  
CMD ["npm","start"]  
  
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   ^U Undo  
^X Exit      ^R Read File  ^N Replace    ^U Paste      ^D Justify    ^G Go To Line ^E Redo
```

```
ubuntu@devops-instance: ~  
Expanded Security Maintenance for Applications is not enabled.  
0 updates can be applied immediately.  
  
1 additional security update can be applied with ESM Apps.  
Learn more about enabling ESM Apps service at https://ubuntu.com/esm  
  
Last login: Thu Mar  5 14:47:07 2026 from 10.20.155.1  
ubuntu@devops-instance:~$ docker --version  
Docker version 28.2.2, build 28.2.2-0ubuntu1-24.04.1  
ubuntu@devops-instance:~$ sudo apt update  
sudo apt install docker.io -y  
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease  
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease  
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease  
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
1 package can be upgraded. Run 'apt list --upgradable' to see it.  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
docker.io is already the newest version (28.2.2-0ubuntu1-24.04.1).  
0 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.  
ubuntu@devops-instance:~$ sudo systemctl start docker  
sudo systemctl enable docker  
ubuntu@devops-instance:~$ docker build -t blitz-app .  
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.  
Install the buildx component to build images with BuildKit:  
https://docs.docker.com/go/buildx/  
  
unable to prepare context: unable to evaluate symlinks in Dockerfile path: lstat /home/ubuntu/Dockerfile  
: no such file or directory  
ubuntu@devops-instance:~$
```



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```
ubuntu@devops-instance: ~/LeetCode-Throne
1 package can be upgraded. Run 'apt list --upgradable' to see it.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
docker.io is already the newest version (28.2.2-0ubuntu1~24.04.1).
0 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.
ubuntu@devops-instance:~$ sudo systemctl start docker
sudo systemctl enable docker
ubuntu@devops-instance:~$ docker build -t blitz-app .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

unable to prepare context: unable to evaluate symlinks in Dockerfile path: lstat /home/ubuntu/Dockerfile
: no such file or directory
ubuntu@devops-instance:~$ cd ~/LeetCode-Throne
ubuntu@devops-instance:~/LeetCode-Throne$ ls
PROJECT_DOCUMENTATION.md  high_level_design.pdf  public          test-image-upload.html
README.md                 index.html             render.yaml    tsconfig.app.json
backend                  package-lock.json      src            tsconfig.json
deploy-package.json       package.json           system-architecture.png  tsconfig.node.json
eslint.config.js          postcss.config.js      tailwind.config.js  vite.config.ts
ubuntu@devops-instance:~/LeetCode-Throne$ docker build -t blitz-app .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

unable to prepare context: unable to evaluate symlinks in Dockerfile path: lstat /home/ubuntu/LeetCode-Throne/Dockerfile: no such file or directory
ubuntu@devops-instance:~/LeetCode-Throne$ nano Dockerfile
ubuntu@devops-instance:~/LeetCode-Throne$ ls
Dockerfile      high_level_design.pdf  render.yaml    tsconfig.json
PROJECT_DOCUMENTATION.md  index.html            src            tsconfig.node.json
README.md       package-lock.json      system-architecture.png  vite.config.ts
backend         package.json           tailwind.config.js
deploy-package.json  postcss.config.js    test-image-upload.html
eslint.config.js    public               tsconfig.app.json
ubuntu@devops-instance:~/LeetCode-Throne$
```

```
ubuntu@devops-instance: ~
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ multipass shell devops-instance
update.go:85: cannot change mount namespace according to change mount (/run/user/1000/doc/by-app/snap.mu
ltipass /run/user/1000/doc none bind,rw,x-snapd.ignore-missing 0 0): cannot inspect "/run/user/1000/doc"
: lstat /run/user/1000/doc: permission denied
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-101-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Mar  5 15:01:40 IST 2026

System load:  0.11               Processes:            372
Usage of /:   11.7% of 18.33GB   Users logged in:     0
Memory usage: 8%                IPv4 address for ens3: 10.20.155.17
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

1 update can be applied immediately.
1 of these updates is a standard security update.
To see these additional updates run: apt list --upgradable

1 additional security update can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Thu Mar  5 15:00:14 2026 from 10.20.155.1
ubuntu@devops-instance:~$ git clone https://github.com/CodingEnthusiastic/LeetCode-Throne.git
Cloning into 'LeetCode-Throne'...
remote: Enumerating objects: 2035, done.
remote: Counting objects: 100% (355/355), done.
remote: Compressing objects: 100% (167/167), done.
remote: Total 2035 (delta 267), reused 228 (delta 184), pack-reused 1680 (from 2)
Receiving objects: 100% (2035/2035), 43.45 MiB | 17.99 MiB/s, done.
Resolving deltas: 100% (1415/1415), done.
ubuntu@devops-instance:~$
```



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```
ubuntu@devops-instance: ~/LeetCode-Throne
unable to prepare context: unable to evaluate symlinks in Dockerfile path: lstat /home/ubuntu/LeetCode-Throne/Dockerfile: no such file or directory
ubuntu@devops-instance:~/LeetCode-Throne$ FROM node:18

WORKDIR /app

COPY package*.json ./

RUN npm install

COPY . .

EXPOSE 3000

CMD ["npm","start"]^C
ubuntu@devops-instance:~/LeetCode-Throne$ nano Dockerfile
ubuntu@devops-instance:~/LeetCode-Throne$ docker build -t blitz-app .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  92.82MB
Step 1/7 : FROM node:18
18: Pulling from library/node
3e6b9d1a9511: Pulling fs layer
37927ed901b1: Pulling fs layer
79b2f47ad444: Pulling fs layer
e23f099911d6: Pulling fs layer
cda7f44f2bdd: Pulling fs layer
c6b30c3f1696: Pulling fs layer
3697be50c98b: Pulling fs layer
461077a72fb7: Pulling fs layer
e23f099911d6: Waiting
cda7f44f2bdd: Waiting
c6b30c3f1696: Waiting
3697be50c98b: Waiting
461077a72fb7: Waiting

could cause a slowdown up to 100x even if nothing is polyfilled. Some versions have web compatibility issues. Please, upgrade your dependencies to the actual version of core-js.
npm warn deprecated highlight.js@9.12.0: Version no longer supported. Upgrade to @latest

added 608 packages, and audited 609 packages in 12s

204 packages are looking for funding
  run `npm fund` for details

15 vulnerabilities (8 moderate, 7 high)

To address issues that do not require attention, run:
  npm audit fix

To address all issues (including breaking changes), run:
  npm audit fix --force

Run `npm audit` for details.
npm notice
npm notice New major version of npm available! 10.8.2 -> 11.11.0
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.11.0
npm notice To update run: npm install -g npm@11.11.0
npm notice
--> Removed intermediate container 668440d67516
--> 08c02e4cf509
Step 5/7 : COPY . .
--> 06457c9c1e5f
Step 6/7 : EXPOSE 3000
--> Running in a9c790e90855
--> Removed intermediate container a9c790e90855
--> e01f9399a832
Step 7/7 : CMD ["npm","start"]
--> Running in f2dcee3e9482
--> Removed intermediate container f2dcee3e9482
--> 6fb555c9c2eb
Successfully built 6fb555c9c2eb
Successfully tagged blitz-app:latest
ubuntu@devops-instance:~/LeetCode-Throne$
```



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```
ubuntu@devops-instance: ~/LeetCode-Throne

To address issues that do not require attention, run:
  npm audit fix

To address all issues (including breaking changes), run:
  npm audit fix --force

Run 'npm audit' for details.
npm notice
npm notice New major version of npm available! 10.8.2 -> 11.11.0
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.11.0
npm notice To update run: npm install -g npm@11.11.0
npm notice
--> Removed intermediate container b21992556133
--> a8c69d1bc665
Step 5/7 : COPY . .
--> 5314cb4f74b2
Step 6/7 : EXPOSE 3000
--> Running in ab38d03434fb
--> Removed intermediate container ab38d03434fb
--> ddd494bac858
Step 7/7 : CMD ["npm","start"]
--> Running in 1c78167c02f4
--> Removed intermediate container 1c78167c02f4
--> 6c93dbccb976
Successfully built 6c93dbccb976
Successfully tagged blitz-app:latest
ubuntu@devops-instance:~/LeetCode-Throne$ docker rm blitz-container
blitz-container
ubuntu@devops-instance:~/LeetCode-Throne$ docker run -d -p 3000:3000 --name blitz-container blitz-app
88a05f6aadd03f491ec9507de9831f30c97a1af10a4d9074e63f01db237c4508
ubuntu@devops-instance:~/LeetCode-Throne$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
88a05f6aadd0   blitz-app      "docker-entrypoint.s..." 9 seconds ago  Up 9 seconds  0.0.0.0:3000->3000/tcp
p, [::]:3000->3000/tcp
ubuntu@devops-instance:~/LeetCode-Throne$ multipass list
multipass: command not found
ubuntu@devops-instance:~/LeetCode-Throne$
```

```
ubuntu@devops-instance: ~/LeetCode-Throne

(!) Some chunks are larger than 500 kB after minification. Consider:
- Using dynamic import() to code-split the application
- Use build.rollupOptions.output.manualChunks to improve chunking: https://rollups.org/configuration-options/#output-manualchunks
- Adjust chunk size limit for this warning via build.chunkSizeWarningLimit.
--> Removed intermediate container 509ae7812672
--> a8d1cb223dba
Step 7/8 : EXPOSE 3000
--> Running in 536d49d7b769
--> Removed intermediate container 536d49d7b769
--> f3243d1eacb0
Step 8/8 : CMD ["npm","run","preview","--","--host","0.0.0.0","--port","3000"]
--> Running in 5ee4ad964727
--> Removed intermediate container 5ee4ad964727
--> 8547e22e4d08
Successfully built 8547e22e4d08
Successfully tagged blitz-app:latest
ubuntu@devops-instance:~/LeetCode-Throne$ docker run -d -p 3000:3000 --name blitz-container blitz-app
7d2c8ee09e0a92606d3a906ae0a02f010d7203bc1a6eb63fa5867fb644b1748e
ubuntu@devops-instance:~/LeetCode-Throne$ curl localhost:3000
<!doctype html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <!-- <link rel="icon" type="image/svg+xml" href="/vite.svg" /> -->
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <link rel="icon" type="image/png" href="https://tse3.mm.bing.net/th?id/OIP.1tecZz8ycQUqoyG091habgHaE8?pid=Api&P=0&h=180" />
  <title>CodeThrone</title>
  <script type="module" crossorigin src="/assets/index-CMnEYOJK.js"></script>
  <link rel="stylesheet" crossorigin href="/assets/index-BGLfvPbS.css">
</head>
<body>
  <div id="root"></div>
</body>
</html>
ubuntu@devops-instance:~/LeetCode-Throne$
```




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```
Activities Terminal Apr 6 10:39
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009: ~
Preparing to unpack .../01-gpg-wks-client_2.2.27-3ubuntu2.5_amd64.deb ...
Unpacking gpg-wks-client (2.2.27-3ubuntu2.5) over (2.2.27-3ubuntu2.1) ...
Preparing to unpack .../02-dirmngr_2.2.27-3ubuntu2.5_amd64.deb ...
Unpacking dirmngr (2.2.27-3ubuntu2.5) over (2.2.27-3ubuntu2.1) ...
Preparing to unpack .../03-gpg-wks-server_2.2.27-3ubuntu2.5_amd64.deb ...
Unpacking gpg-wks-server (2.2.27-3ubuntu2.5) over (2.2.27-3ubuntu2.1) ...
Preparing to unpack .../04-gnupg-utils_2.2.27-3ubuntu2.5_amd64.deb ...
Unpacking gnupg-utils (2.2.27-3ubuntu2.5) over (2.2.27-3ubuntu2.1) ...
Preparing to unpack .../05-gpg-agent_2.2.27-3ubuntu2.5_amd64.deb ...
Unpacking gpg-agent (2.2.27-3ubuntu2.5) over (2.2.27-3ubuntu2.1) ...
Preparing to unpack .../06-gpg_2.2.27-3ubuntu2.5_amd64.deb ...
Unpacking gpg (2.2.27-3ubuntu2.5) over (2.2.27-3ubuntu2.1) ...
Preparing to unpack .../07-gpgconf_2.2.27-3ubuntu2.5_amd64.deb ...
Unpacking gpgconf (2.2.27-3ubuntu2.5) over (2.2.27-3ubuntu2.1) ...
Preparing to unpack .../08-gnupg-l10n_2.2.27-3ubuntu2.5_all.deb ...
Unpacking gnupg-l10n (2.2.27-3ubuntu2.5) over (2.2.27-3ubuntu2.1) ...
Preparing to unpack .../09-gnupg_2.2.27-3ubuntu2.5_all.deb ...
Unpacking gnupg (2.2.27-3ubuntu2.5) over (2.2.27-3ubuntu2.1) ...
Preparing to unpack .../10-gpgsm_2.2.27-3ubuntu2.5_amd64.deb ...
Unpacking gpgsm (2.2.27-3ubuntu2.5) over (2.2.27-3ubuntu2.1) ...
Preparing to unpack .../11-libsqlite3-0_3.37.2-2ubuntu0.5_amd64.deb ...
Unpacking libsqlite3-0:amd64 (3.37.2-2ubuntu0.5) over (3.37.2-2ubuntu0.3) ...
Preparing to unpack .../12-gpgv_2.2.27-3ubuntu2.5_amd64.deb ...
Unpacking gpgv (2.2.27-3ubuntu2.5) over (2.2.27-3ubuntu2.1) ...
Setting up gpgv (2.2.27-3ubuntu2.5) ...
(Reading database ... 201581 files and directories currently installed.)
Preparing to unpack .../000-libpython3.10_3.10.12-1-22.04.15_amd64.deb ...
Unpacking libpython3.10:amd64 (3.10.12-1-22.04.15) over (3.10.12-1-22.04.5) ...
Preparing to unpack .../001-python3.10_3.10.12-1-22.04.15_amd64.deb ...
Unpacking python3.10 (3.10.12-1-22.04.15) over (3.10.12-1-22.04.5) ...
Preparing to unpack .../002-libpython3.10-stdlib_3.10.12-1-22.04.15_amd64.deb ...
Unpacking libpython3.10-stdlib:amd64 (3.10.12-1-22.04.15) over (3.10.12-1-22.04.5) ...
Preparing to unpack .../003-python3.10-minimal_3.10.12-1-22.04.15_amd64.deb ...
Unpacking python3.10-minimal (3.10.12-1-22.04.15) over (3.10.12-1-22.04.5) ...
Preparing to unpack .../004-libpython3.10-minimal_3.10.12-1-22.04.15_amd64.deb ...
Unpacking libpython3.10-minimal:amd64 (3.10.12-1-22.04.15) over (3.10.12-1-22.04.5) ...
Progress: [ 15%] [#####.....]
```

```
Activities Terminal Apr 6 10:44
ubuntu@devops-instance: ~/LeetCode-Throne
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009: $ multipass shell devops-instance
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-101-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Apr  6 10:43:26 IST 2026

System load:  0.06          Processes:      119
Usage of /:   26.0% of 18.3GB Users logged in:  0
Memory usage: 14%          IPv4 address for ens3: 10.20.155.17
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.
1 additional security update can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Thu Mar  5 15:01:40 2026 from 10.20.155.1
ubuntu@devops-instance:~$ docker --version
docker ps
Docker version 28.2.2, build 28.2.2-0ubuntu1-24.04.1
CONTAINER ID   IMAGE      COMMAND                  CREATED    STATUS    PORTS
7d2c8ee09e0a   blitz-app "docker-entrypoint.s..." 4 weeks ago Up 4 weeks 0.0.0.0:3000->3000/tcp, [
::]:3000->3000/tcp blitz-container
ubuntu@devops-instance:~$ cd ~/LeetCode-Throne
docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED    STATUS    PORTS
7d2c8ee09e0a   blitz-app "docker-entrypoint.s..." 4 weeks ago Up 4 weeks 0.0.0.0:3000->3000/tcp, [
::]:3000->3000/tcp blitz-container
ubuntu@devops-instance:~/LeetCode-Throne$
```



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```
Activities Terminal Apr 6 10:45
ubuntu@devops-instance: ~/LeetCode-Throne
1 additional security update can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Thu Mar 5 15:01:40 2026 from 10.20.155.1
ubuntu@devops-instance:~$ docker --version
Docker version 28.2.2, build 28.2.2-0ubuntu1-24.04.1
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS
7d2c8ee09e0a   blitz-app "docker-entrypoint.s..." 4 weeks ago   Up 4 weeks   0.0.0.0:3000->3000/tcp, [
::]:3000->3000/tcp   blitz-container
ubuntu@devops-instance:~$ cd ~/LeetCode-Throne
docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS
7d2c8ee09e0a   blitz-app "docker-entrypoint.s..." 4 weeks ago   Up 4 weeks   0.0.0.0:3000->3000/tcp, [
::]:3000->3000/tcp   blitz-container
ubuntu@devops-instance:~/LeetCode-Throne$ docker start blitz-container
blitz-container
ubuntu@devops-instance:~/LeetCode-Throne$ curl localhost:3000
<doctype html>
<html lang="en">
<head>
<meta charset="UTF-8" />
<!-- <link rel="icon" type="image/svg+xml" href="/vite.svg" /> -->
<meta name="viewport" content="width=device-width, initial-scale=1.0" />
<link rel="icon" type="image/png" href="https://tse3.mm.bing.net/th/id/OIP.1tecZz8ycQUqoyG091habgHaE
8?pid=Api&P=0&h=180" />
<title>CodeThrone</title>
<script type="module" crossorigin src="/assets/index-CMnEYOJK.js"></script>
<link rel="stylesheet" crossorigin href="/assets/index-BGLfvPbs.css">
</head>
<body>
<div id="root"></div>
</body>
</html>
ubuntu@devops-instance:~/LeetCode-Throne$
```

Git

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Code

About

CodeThrone is a LeetCode-style coding practice platform designed to help users strengthen problem-solving skills through curated challenges. It features algorithmic problems, test case validation, game feature, discord feature and instant feedback, enabling learners to practice, debug, and optimize solutions.

codethrone.netlify.app/

socket webirc video-processing mem-stack

Readme

Activity

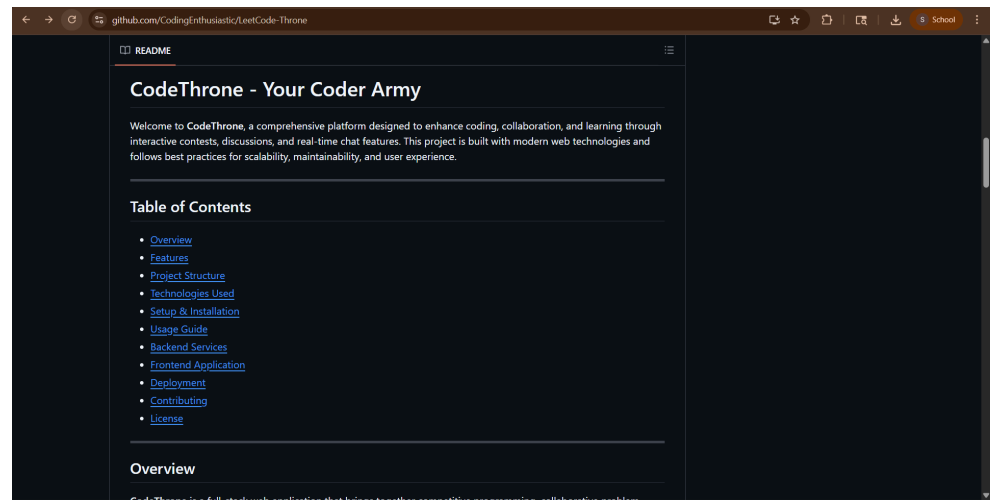
7 stars

0 watching

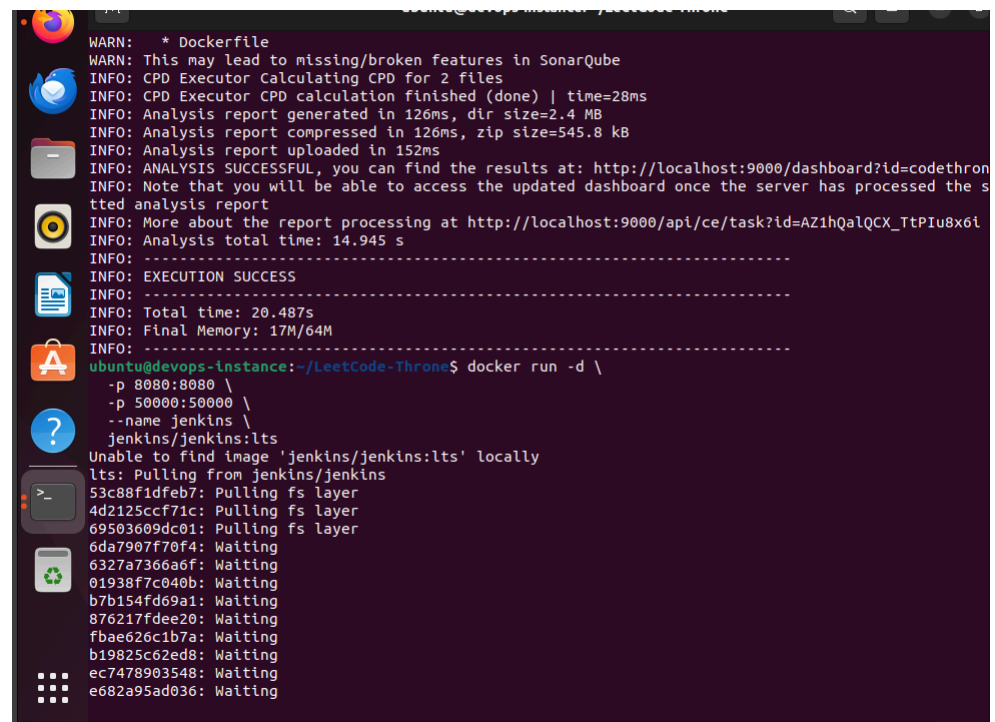
Commit	Message	Author	Date
b41c585	Change agent to Docker with Node.js image	b41c585	last month
...	...	...	...



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Jenkins

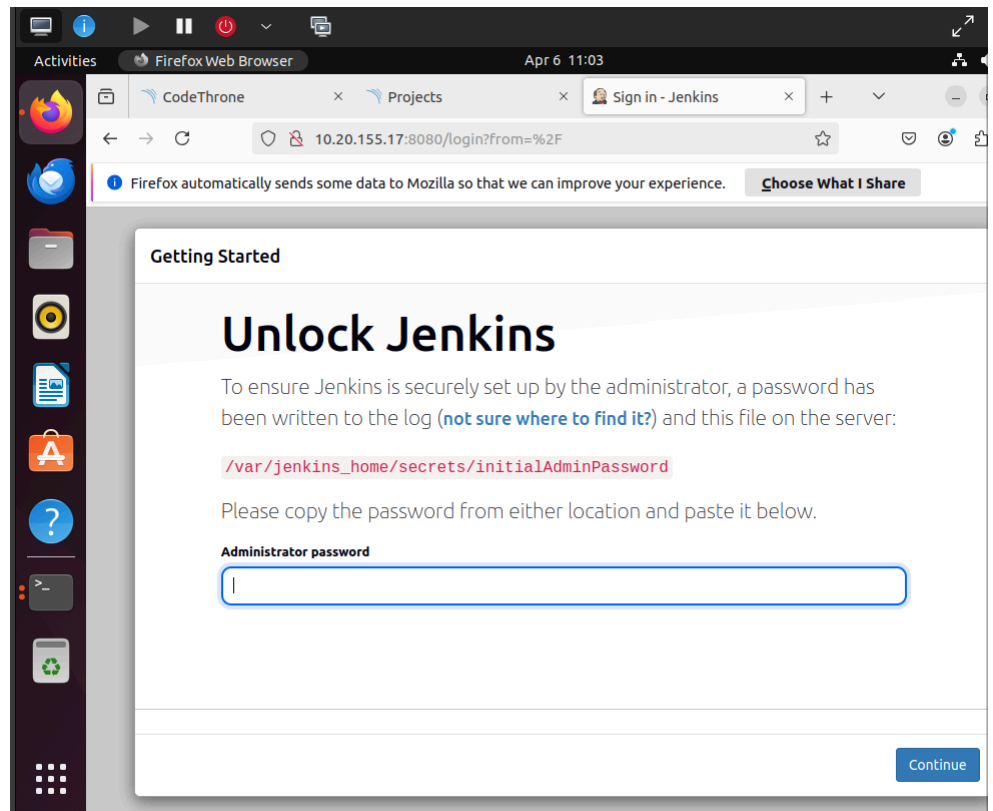




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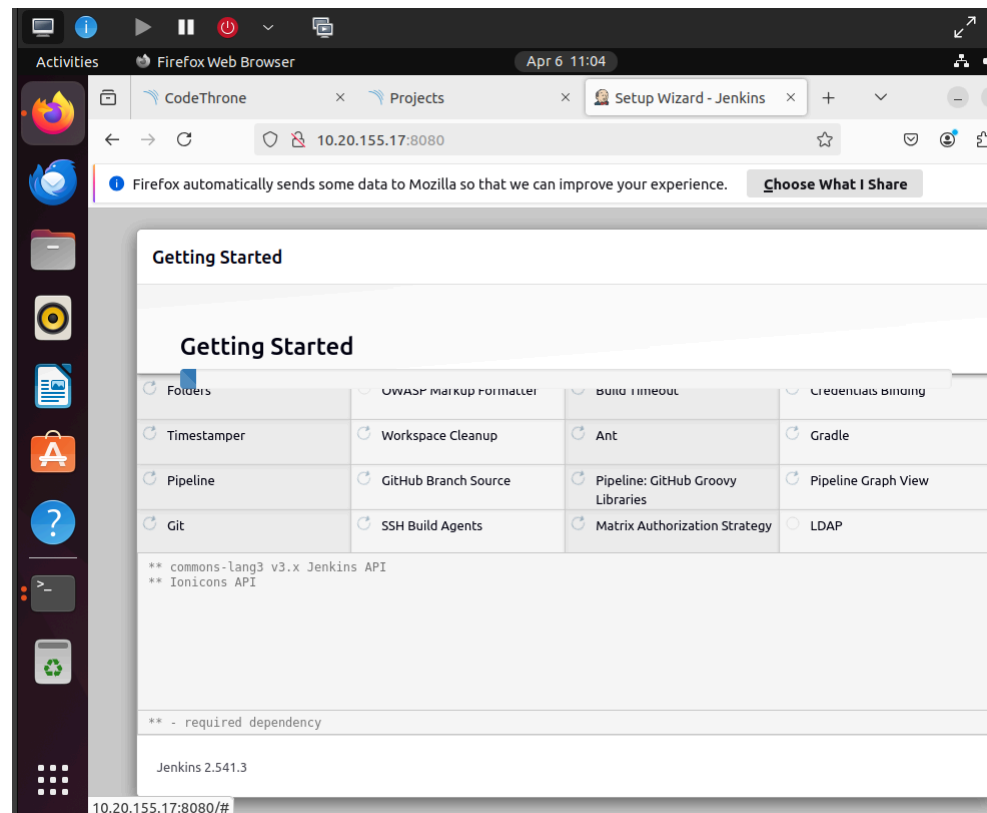
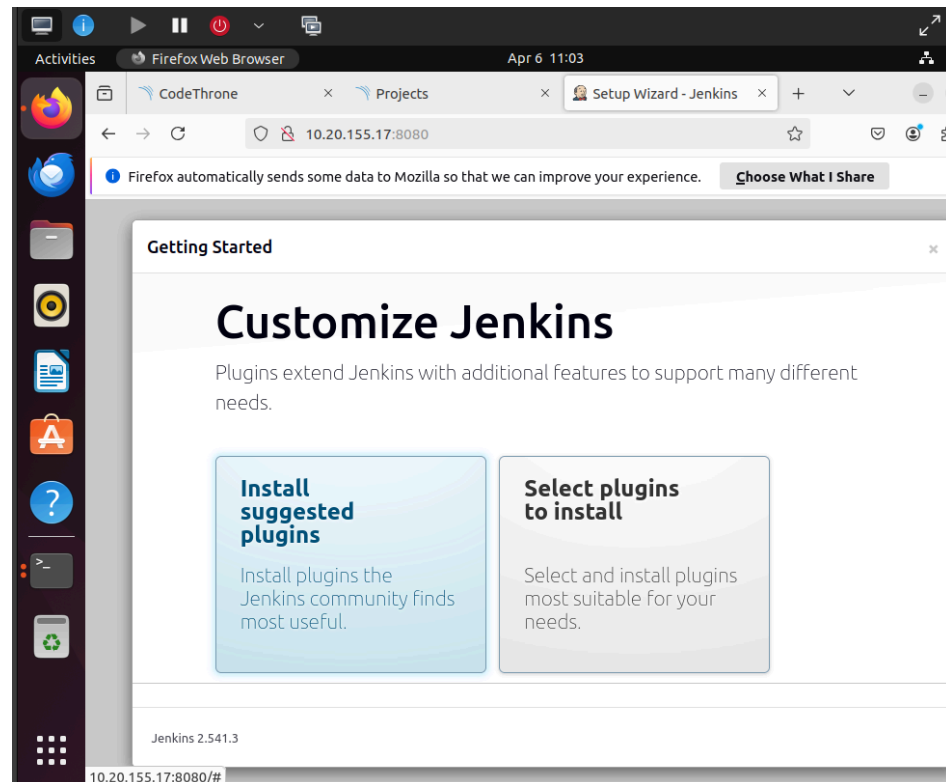
```
ubuntu@devops-instance: ~/LeetCode-Throne
INFO: CPD Executor CPD calculation finished (done) | time=28ms
INFO: Analysis report generated in 126ms, dir size=2.4 MB
INFO: Analysis report compressed in 126ms, zip size=545.8 kB
INFO: Analysis report uploaded in 152ms
INFO: ANALYSIS SUCCESSFUL, you can find the results at: http://localhost:9000/dashboard?id=codethron
INFO: Note that you will be able to access the updated dashboard once the server has processed the s
tted analysis report
INFO: More about the report processing at http://localhost:9000/api/ce/task?id=AZ1hQalQCX_TtPiU8x6t
INFO: Analysis total time: 14.945 s
INFO: -----
INFO: EXECUTION SUCCESS
INFO: -----
INFO: Total time: 20.487s
INFO: Final Memory: 17M/64M
INFO: -----
ubuntu@devops-instance:~/LeetCode-Throne$ docker run -d \
  -p 8080:8080 \
  -p 50000:50000 \
  --name jenkins \
  jenkins/jenkins:lts
Unable to find image 'jenkins/jenkins:lts' locally
lts: Pulling from jenkins/jenkins
53c88f1dfeb7: Pull complete
4d2125ccf71c: Pull complete
69503609dc01: Pull complete
6da7907f70f4: Pull complete
6327a7366a6f: Pull complete
01938f7c040b: Pull complete
b7b154fd69a1: Pull complete
876217fdee20: Pull complete
fbae626c1b7a: Pull complete
b19825c62ed8: Pull complete
ec7478903548: Pull complete
e682a95ad036: Pull complete
Digest: sha256:c4098086090ca98491d4bf66182f5e3b015a8232f2acf2df209a212a5801aa8e
Status: Downloaded newer image for jenkins/jenkins:lts
ec9191e95b30553513fa04d4ee4eb577293c851a7169c54e2409d91ea4d7a9bb
ubuntu@devops-instance:~/LeetCode-Throne$
```

Password : 5d6c1910071b4a26b1aa38c6521263b2



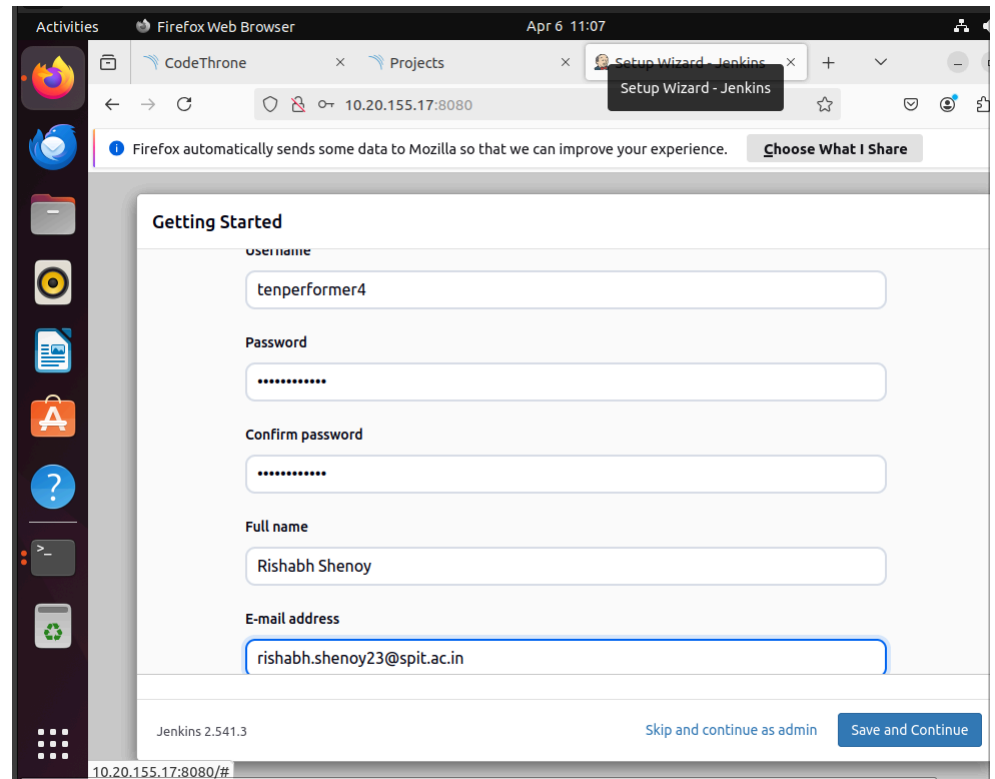
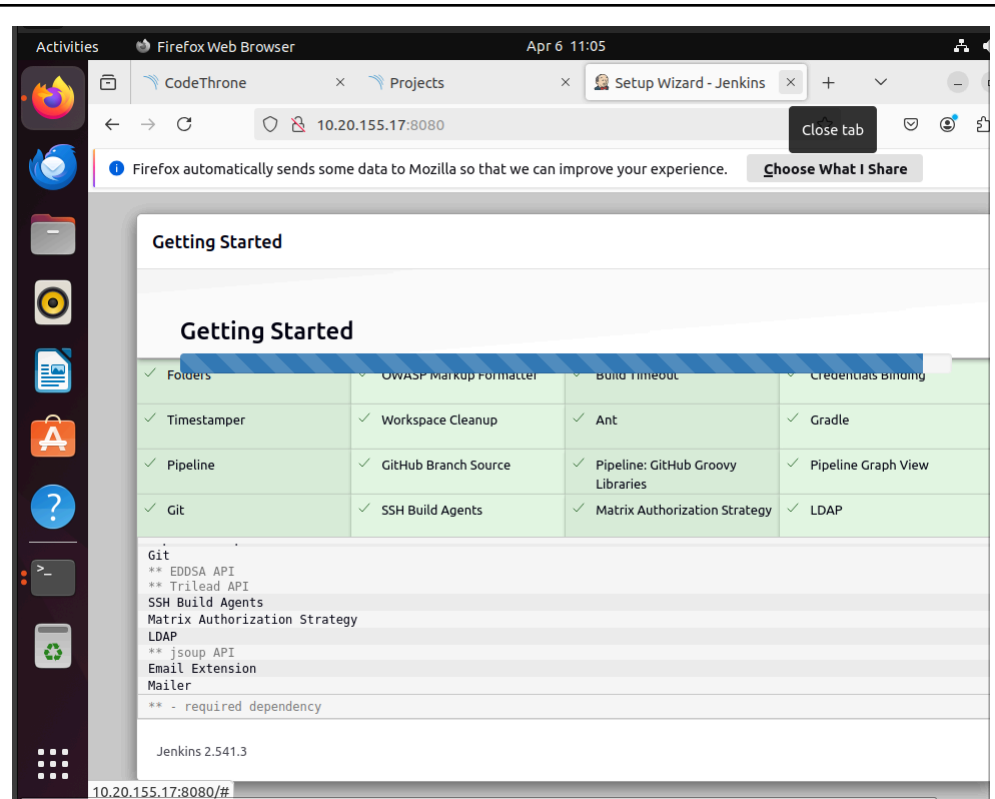


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Jenkins URL : <http://10.20.155.17:8080/>



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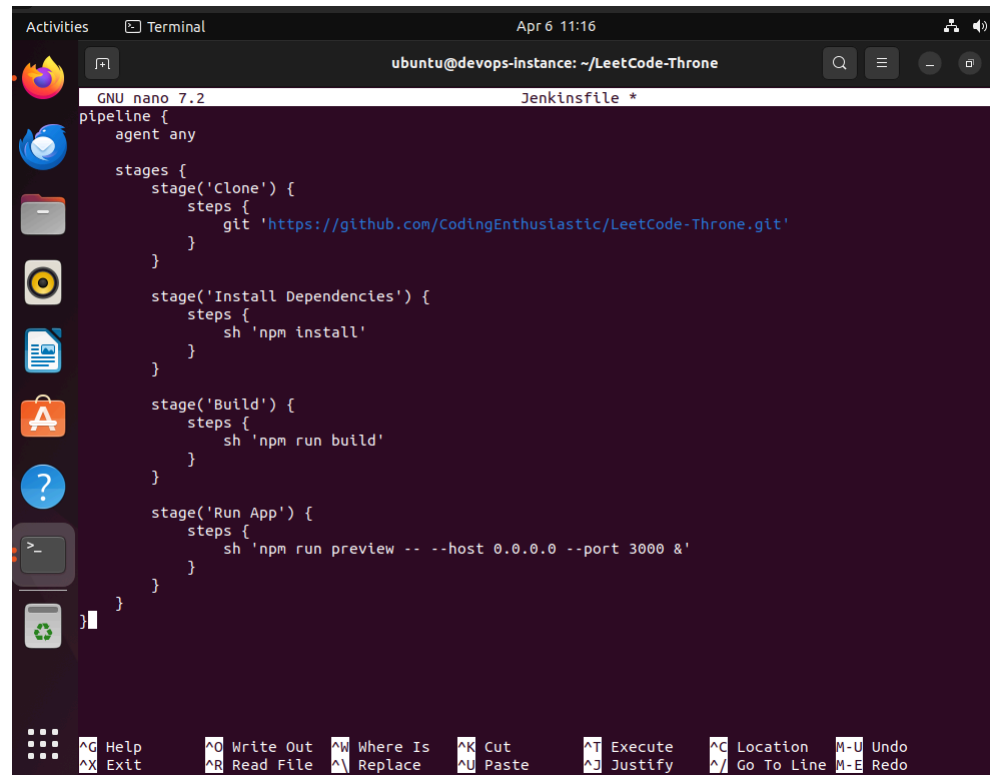
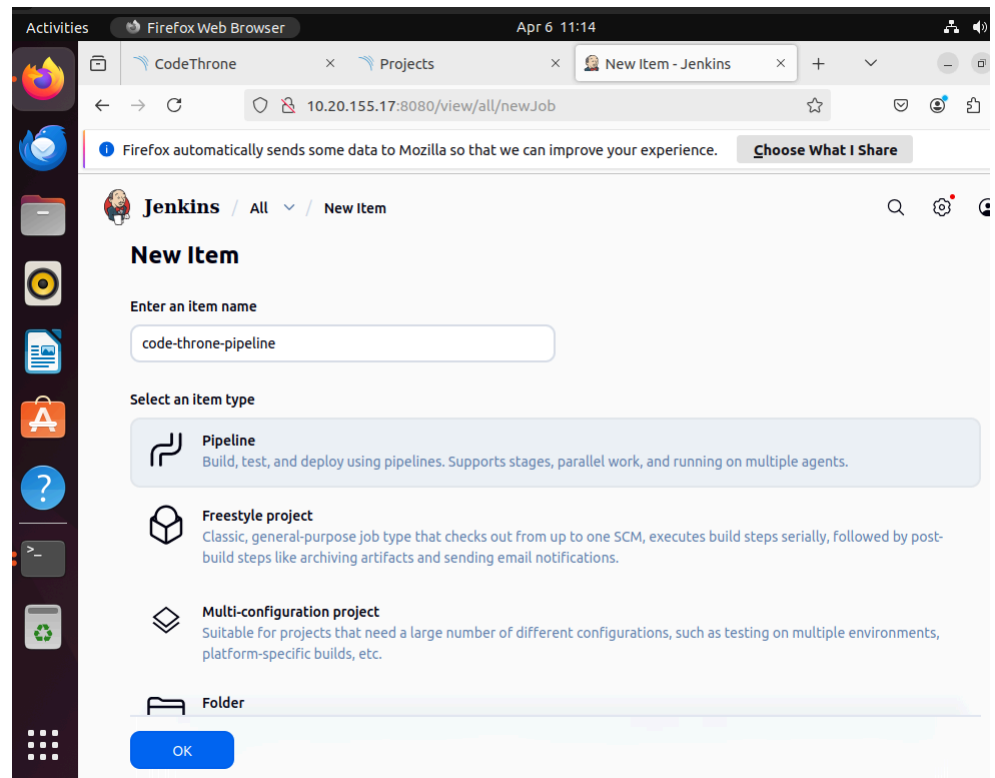
The image displays two sequential screenshots of the Jenkins Setup Wizard interface, accessed via a Firefox Web Browser. The browser's address bar shows the URL `10.20.155.17:8080`.

Top Screenshot (Apr 6 11:07): The page is titled "Getting Started" and "Instance Configuration". It prompts the user to enter the "Jenkins URL". The input field contains `http://10.20.155.17:8080/`. Below the input field, explanatory text states: "The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the BUILD_URL environment variable provided to build steps. The proposed default value shown is **not saved yet** and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links." At the bottom right, there are two buttons: "Not now" and "Save and Finish".

Bottom Screenshot (Apr 6 11:08): The page is titled "Getting Started" and "Jenkins is ready!". It displays the message "Your Jenkins setup is complete." and a blue button labeled "Start using Jenkins".



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Activities Firefox Web Browser Apr 6 11:30

CodeThrone Projects Download progress Docker | Jenkins | +

10.20.155.17:8080/manage/pluginManager/updates/

Firefox automatically sends some data to Mozilla so that we can improve your experience. Choose What I Share

Jenkins / Manage Jenkins / Plugins

Plugins

- Updates
- Available plugins
- Installed plugins
- Advanced settings
- Download progress**

Download progress

Preparation

- Checking internet connectivity
- Checking update center connectivity
- Success

commons-lang3 v3.x Jenkins API	Success
Ionicons API	Success
Folders	Success
OWASP Markup Formatter	Success
ASM API	Success
JSON Path API	Success
Structs	Success
Pipeline: Step API	Success
commons-text API	Success
Token Macro	Success
Build Timeout	Success
bouncycastle API	Success
Credentials	Success

Activities Terminal Apr 6 11:41

ubuntu@devops-instance: ~/LeetCode-Throne

```
Hit:7 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
W: GPG error: https://pkg.jenkins.io/debian-stable binary/ Release: The following signatures couldn't
verified because the public key is not available: NO_PUBKEY 7198F4B714ABFC68
E: The repository 'https://pkg.jenkins.io/debian-stable binary/ Release' is not signed.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package jenkins is not available, but is referred to by another package.
This may mean that the package is missing, has been obsoleted, or
is only available from another source
E: Package 'jenkins' has no installation candidate
ubuntu@devops-instance:~/LeetCode-Throne$ sudo rm -f /usr/share/keyrings/jenkins-keyring.asc
sudo rm -f /etc/apt/sources.list.d/jenkins.list
ubuntu@devops-instance:~/LeetCode-Throne$ curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-
3.key | sudo gpg --dearmor -o /usr/share/keyrings/jenkins-keyring.gpg
ubuntu@devops-instance:~/LeetCode-Throne$ echo "deb [signed-by=/usr/share/keyrings/jenkins-keyring.gp
https://pkg.jenkins.io/debian-stable binary/" | sudo tee /etc/apt/sources.list.d/jenkins.list
deb [signed-by=/usr/share/keyrings/jenkins-keyring.gpg] https://pkg.jenkins.io/debian-stable binary/
ubuntu@devops-instance:~/LeetCode-Throne$ sudo apt update
Ign:1 https://pkg.jenkins.io/debian-stable binary/ InRelease
Get:2 https://pkg.jenkins.io/debian-stable binary/ Release [2044 B]
Get:3 https://pkg.jenkins.io/debian-stable binary/ Release.gpg [833 B]
Ign:3 https://pkg.jenkins.io/debian-stable binary/ Release.gpg
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:6 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:7 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
W: GPG error: https://pkg.jenkins.io/debian-stable binary/ Release: The following signatures couldn't
verified because the public key is not available: NO_PUBKEY 7198F4B714ABFC68
E: The repository 'https://pkg.jenkins.io/debian-stable binary/ Release' is not signed.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
ubuntu@devops-instance:~/LeetCode-Throne$
```




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```
vities Terminal Apr 6 11:57
ubuntu@devops-instance: ~/LeetCode-Throne

2026-04-06 06:20:34.474+0000 [id=32] INFO winstone.Logger#logInternal: JVM is terminating. Shutting
down Jetty
2026-04-06 06:20:34.474+0000 [id=32] INFO org.eclipse.jetty.server.Server#doStop: Stopped oejs.Ser
ver@53a665ad[STOPPING][12.1.5,sto=0]
2026-04-06 06:20:34.476+0000 [id=32] INFO o.e.j.server.AbstractConnector#doStop: Stopped oejs.Serv
erConnector@2aa6311a[HTTP/1.1, (http/1.1)][0.0.0.0:8080]
2026-04-06 06:20:34.489+0000 [id=32] INFO winstone.Logger#logInternal: Jetty shutdown successfully
ubuntu@devops-instance:~/LeetCode-Throne$ docker ps
CONTAINER ID   IMAGE                                COMMAND                                  CREATED        STATUS
PORTS
07ea7007e07e   sonarqube:lts-community             "/opt/sonarqube/dock..." About an hour ago Up About an hour
0.0.0.0:9000->9000/tcp, [::]:9000->9000/tcp sonarqube
7d2c8ee09e0a   blitz-app                           "docker-entrypoint.s..." 4 weeks ago    Up 4 weeks
0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp blitz-container
ubuntu@devops-instance:~/LeetCode-Throne$ docker start jenkins
jenkins
ubuntu@devops-instance:~/LeetCode-Throne$ docker ps
CONTAINER ID   IMAGE                                COMMAND                                  CREATED        STATUS        NAMES
PORTS
8ddaff954546   jenkins/jenkins:lts                "/usr/bin/tini -- /u..." 10 minutes ago Up 3 seconds   jenkins
0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp
07ea7007e07e   sonarqube:lts-community             "/opt/sonarqube/dock..." About an hour ago Up About an hour sonarqub
0.0.0.0:9000->9000/tcp, [::]:9000->9000/tcp
e
7d2c8ee09e0a   blitz-app                           "docker-entrypoint.s..." 4 weeks ago    Up 4 weeks    blitz-co
0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
ntainer
ubuntu@devops-instance:~/LeetCode-Throne$ docker rm -f jenkins
jenkins
ubuntu@devops-instance:~/LeetCode-Throne$ docker run -d \
-p 8080:8080 \
-p 50000:50000 \
-v jenkins_home:/var/jenkins_home \
-v /var/run/docker.sock:/var/run/docker.sock \
--name jenkins \
jenkins/jenkins:lts
fa4ea62ad2ed0863c4f981289bca38b91d2ddaedd4fd186f7da155a03ee92fae
ubuntu@devops-instance:~/LeetCode-Throne$
```

```
vities Terminal Apr 6 11:59
ubuntu@devops-instance: ~/LeetCode-Throne

er.socket'.
Setting up dbus-user-session (1.16.2-2) ...
Setting up criu (4.1.1-1) ...
Setting up python3-protobuf (3.21.12-11) ...
Setting up python3-pycru (4.1.1-1) ...
Setting up dmsetup (2:1.02.205-2) ...
Setting up libdevmapper1.02.1:amd64 (2:1.02.205-2) ...
Setting up libcryptsetup12:amd64 (2:2.7.5-2) ...
Setting up systemd-cryptsetup (257.9-1-deb13u1) ...
Processing triggers for procps (2:4.0.4-9) ...
grep: /proc/1/environ: Permission denied
procps: Applying updated sysctl configuration
sysctl: permission denied on key "kernel.core_pattern"
sysctl: permission denied on key "kernel.sysrq"
sysctl: permission denied on key "kernel.core_uses_pid"
sysctl: permission denied on key "net.ipv4.conf.default.rp_filter"
sysctl: permission denied on key "net.ipv4.conf.eth0.rp_filter"
sysctl: permission denied on key "net.ipv4.conf.lo.rp_filter"
sysctl: permission denied on key "net.ipv4.conf.default.accept_source_route"
sysctl: permission denied on key "net.ipv4.conf.eth0.accept_source_route"
sysctl: permission denied on key "net.ipv4.conf.lo.accept_source_route"
sysctl: permission denied on key "net.ipv4.conf.default.promote_secondaries"
sysctl: permission denied on key "net.ipv4.conf.eth0.promote_secondaries"
sysctl: permission denied on key "net.ipv4.conf.lo.promote_secondaries"
sysctl: permission denied on key "net.ipv4.ping_group_range", ignoring
sysctl: permission denied on key "fs.protected_hardlinks"
sysctl: permission denied on key "fs.protected_symlinks"
sysctl: permission denied on key "fs.protected_regular"
sysctl: permission denied on key "fs.protected_fifos"
sysctl: permission denied on key "vm.max_map_count"
sysctl: permission denied on key "kernel.pid_max"
Processing triggers for libc-bin (2.41-12) ...
Processing triggers for systemd (257.9-1-deb13u1) ...
root@fa4ea62ad2ed:/# exit
exit
ubuntu@devops-instance:~/LeetCode-Throne$ docker exec -it jenkins docker --version
Docker version 26.1.5+dfsg1, build a72d7cd
ubuntu@devops-instance:~/LeetCode-Throne$
```



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Firefox Web Browser

Apr 6 12:02

CodeThrone Projects Problem load Docker | Jenki code-throne x

10.20.155.17:8080/job/code-throne-pipeline/3/

Firefox automatically sends some data to Mozilla so that we can improve your experience. Choose What I Share

Jenkins / code-throne-pipeline #3

Status

</> Changes

Console Output

Edit Build Information

Timings

Git Build Data

Pipeline Overview

Thread Dump

Pause/resume

Replay

Pipeline Steps

Workspaces

Previous Build

#3 (Apr 6, 2026, 6:31:51 AM)

Progress:

Add description Keep this build forever

Started 24 sec ago

Build has been executing for 24 sec

Started by user Rishabh Shenoy

This run spent 21 ms waiting in the queue.

Revision: b40c58512839736c7429727475ea7fbd2015337c

Repository: <https://github.com/CodingEnthusiastic/LeetCode-Throne.git>

refs/remotes/origin/main

No changes.

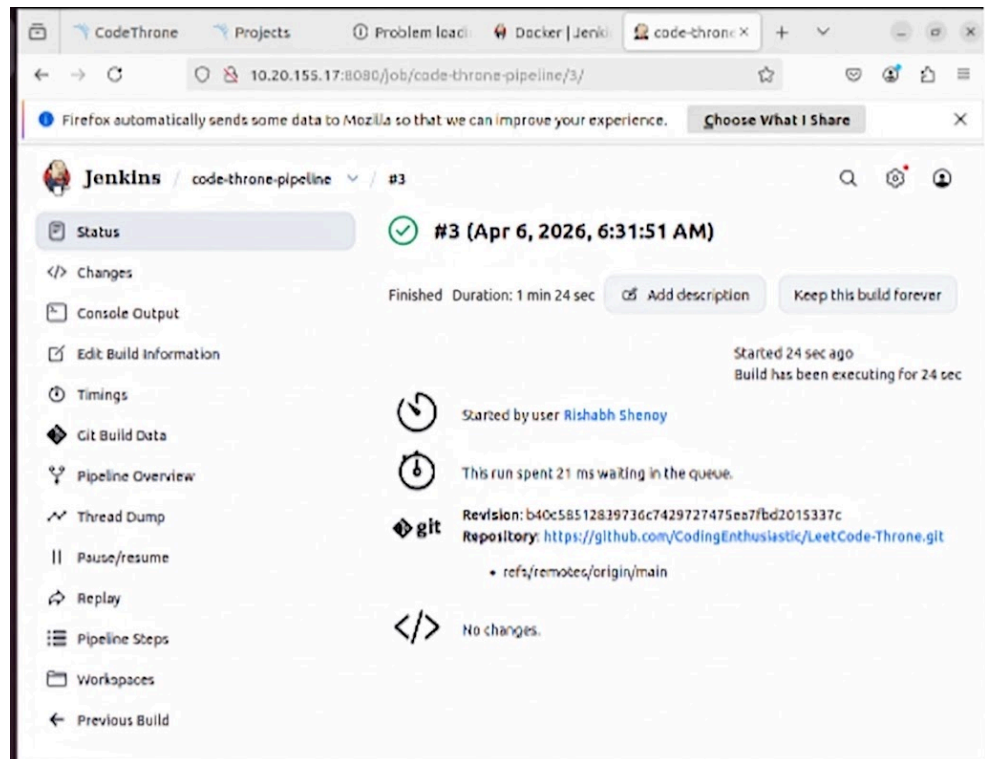
```
i1j:50802->50000/tcp jenkins-docker
mca@mca-ThinkCentre-neo-50s-Gen-3: $ docker stop f484f2c441ae
f484f2c441ae
mca@mca-ThinkCentre-neo-50s-Gen-3: $ docker run -d -p 8080:8080 -p 50000:50000 --name jenkins jenkins/jenkins:lts
docker: Error response from daemon: Conflict. The container name "/jenkins" is already in use by container "e25fb235d33456a81bb3f793270db87a0b706d673c9d2f8aa15b84910375f560". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
mca@mca-ThinkCentre-neo-50s-Gen-3: $ docker rm f484f2c441ae
f484f2c441ae
mca@mca-ThinkCentre-neo-50s-Gen-3: $ docker run -d -p 8080:8080 -p 50000:50000 --name jenkins jenkins/jenkins:lts
docker: Error response from daemon: Conflict. The container name "/jenkins" is already in use by container "e25fb235d33456a81bb3f793270db87a0b706d673c9d2f8aa15b84910375f560". You have to remove (or rename) that container to be able to reuse that name.

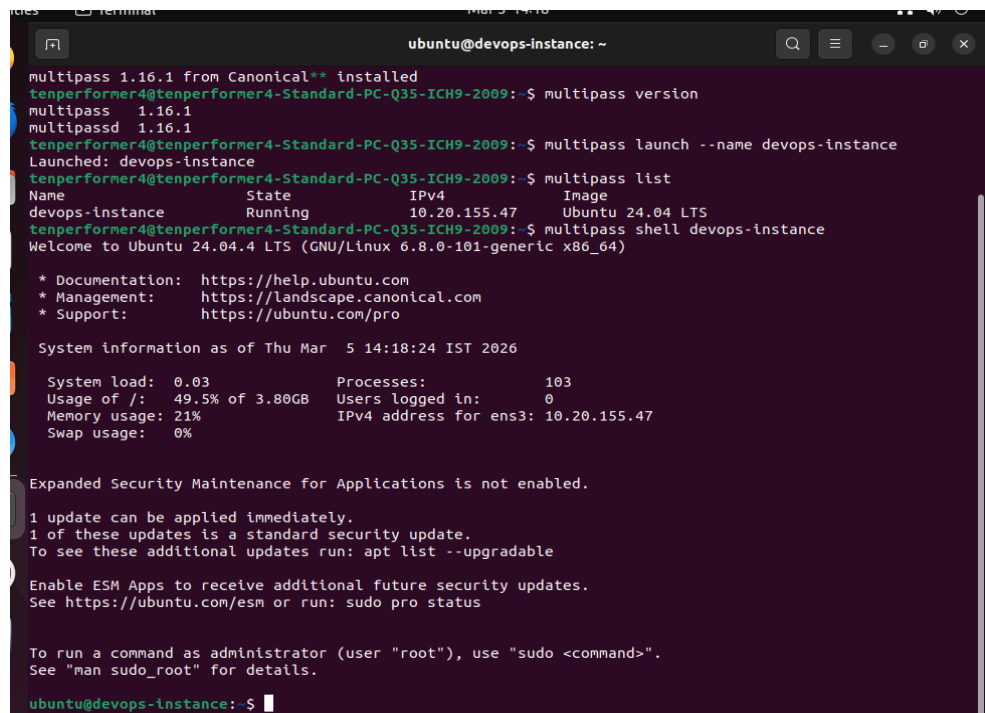
Run 'docker run --help' for more information
mca@mca-ThinkCentre-neo-50s-Gen-3: $ docker rm jenkins
jenkins
mca@mca-ThinkCentre-neo-50s-Gen-3: $ docker run -d -p 8080:8080 -p 50000:50000 --name jenkins jenkins/jenkins:lts
040742866d4c3905adfe634c3db9ac2b8104a03a137127c4bf46f0e9780df6
mca@mca-ThinkCentre-neo-50s-Gen-3: $ docker exec jenkins cat /var/jenkins_home/secrets/initialAdminPassword
bdbc4425337a4448a3f6523c34cccfbf
mca@mca-ThinkCentre-neo-50s-Gen-3: $ ^C
mca@mca-ThinkCentre-neo-50s-Gen-3: $ docker exec jenkins cat /var/jenkins_home/secrets/initialAdminPassword^C
mca@mca-ThinkCentre-neo-50s-Gen-3: $ docker exec jenkins cat /var/jenkins_home/secrets/initialAdminPassword
bdbc4425337a4448a3f6523c34cccfbf
mca@mca-ThinkCentre-neo-50s-Gen-3: $ docker exec -it jenkins bash
jenkins@040742866d4c:/# apt update
apt install maven -y
Error: List directory /var/lib/apt/lists/partial is missing. - Acquire (13: Permission denied)
Error: Could not open lock file /var/lib/dpkg/lock-frontends - open (13: Permission denied)
Error: Unable to acquire the dpkg frontend lock (/var/lib/dpkg/lock-frontends), are you root?
jenkins@040742866d4c:/# exit
exit
mca@mca-ThinkCentre-neo-50s-Gen-3: $ docker exec -u 0 -it jenkins bash
root@040742866d4c:/# apt update
apt install maven -y
Get:1 http://deb.debian.org/debian trixie InRelease [140 kB]
Get:2 http://deb.debian.org/debian trixie-updates InRelease [47.3 kB]
Get:3 http://deb.debian.org/debian-security trixie-security InRelease [43.4 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 Packages [9071 kB]
Get:5 http://deb.debian.org/debian trixie-updates/main amd64 Packages [5412 B]
Get:6 http://deb.debian.org/debian-security trixie-security/main amd64 Packages [123 kB]
Fetched 10.0 MB in 1s (9282 kB/s)
17 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... OK
```



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Multipass & LXD





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```
ubuntu@devops-instance: ~  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
2 packages can be upgraded. Run 'apt list --upgradable' to see them.  
ubuntu@devops-instance:~$ sudo apt install git -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
git is already the newest version (1:2.43.0-1ubuntu7.3).  
git set to manually installed.  
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.  
ubuntu@devops-instance:~$ git --version  
git version 2.43.0  
ubuntu@devops-instance:~$ sudo apt install docker.io -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  bridge-utils containerd dns-root-data dnsmasq-base pigz runc ubuntu-fan  
Suggested packages:  
  ifupdown aufs-tools cgroupfs-mount | cgroup-lite debootstrap docker-buildx docker-compose-v2  
  docker-doc rinse zfs-fuse | zfsutils  
The following NEW packages will be installed:  
  bridge-utils containerd dns-root-data dnsmasq-base docker.io pigz runc ubuntu-fan  
0 upgraded, 8 newly installed, 0 to remove and 2 not upgraded.  
Need to get 76.1 MB of archives.  
After this operation, 288 MB of additional disk space will be used.  
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 pigz amd64 2.8-1 [65.6 kB]  
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 bridge-utils amd64 1.7.1-1ubuntu2 [33.9 kB]  
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 runc amd64 1.3.3-0ubuntu1-24.04.3 [8815  
kB]  
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 containerd amd64 1.7.28-0ubuntu1-24.04.2  
[38.4 MB]  
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 dns-root-data all 2024071801-ubuntu0.24.  
04.1 [5918 B]  
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 dnsmasq-base amd64 2.90-2ubuntu0.1 [376  
kB]  
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 docker.io amd64 28.2.2-0ubuntu1-24.0  
4.1 [28.3 MB]  
Selecting previously unselected package ubuntu-fan.  
Preparing to unpack .../7-ubuntu-fan_0.12.16+24.04.1_all.deb ...  
Unpacking ubuntu-fan (0.12.16+24.04.1) ...  
Setting up dnsmasq-base (2.90-2ubuntu0.1) ...  
Setting up runc (1.3.3-0ubuntu1-24.04.3) ...  
Setting up dns-root-data (2024071801-ubuntu0.24.04.1) ...  
Setting up bridge-utils (1.7.1-1ubuntu2) ...  
Setting up pigz (2.8-1) ...  
Setting up containerd (1.7.28-0ubuntu1-24.04.2) ...  
Created symlink /etc/systemd/system/multi-user.target.wants/containerd.service → /usr/lib/systemd/system  
/containerd.service.  
Setting up ubuntu-fan (0.12.16+24.04.1) ...  
Created symlink /etc/systemd/system/multi-user.target.wants/ubuntu-fan.service → /usr/lib/systemd/system  
/ubuntu-fan.service.  
Setting up docker.io (28.2.2-0ubuntu1-24.04.1) ...  
info: Selecting GID from range 100 to 999 ...  
info: Adding group 'docker' (GID 112) ...  
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/doc  
ker.service.  
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/docker.  
socket.  
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...  
Processing triggers for man-db (2.12.0-4build2) ...  
Scanning processes...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
ubuntu@devops-instance:~$ sudo systemctl start docker  
ubuntu@devops-instance:~$ sudo systemctl enable docker  
ubuntu@devops-instance:~$
```



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```
Setting up dnsmasq-base (2.90-2ubuntu0.1) ...
Setting up runc (1.3.3-0ubuntu1-24.04.3) ...
Setting up dns-root-data (2024071801-ubuntu0.24.04.1) ...
Setting up bridge-utils (1.7.1-1ubuntu2) ...
Setting up pigz (2.8-1) ...
Setting up containerd (1.7.28-0ubuntu1-24.04.2) ...
Created symlink /etc/systemd/system/multi-user.target.wants/containerd.service → /usr/lib/systemd/system/containerd.service.
Setting up ubuntu-fan (0.12.16+24.04.1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/ubuntu-fan.service → /usr/lib/systemd/system/ubuntu-fan.service.
Setting up docker.io (28.2.2-0ubuntu1-24.04.1) ...
info: Selecting GID from range 100 to 999 ...
info: Adding group 'docker' (GID 112) ...
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/docker.socket.
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@devops-instance:~$ sudo systemctl start docker
ubuntu@devops-instance:~$ sudo systemctl enable docker
ubuntu@devops-instance:~$ exit
logout
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ sudo snap install lxd
Download snap "lxd" (37923) from channel "5.21/stable" 78% 17.4MB/s 1.57s
```

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@devops-instance:~$ sudo systemctl start docker
ubuntu@devops-instance:~$ sudo systemctl enable docker
ubuntu@devops-instance:~$ exit
logout
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ sudo snap install lxd
lxd (5.21/stable) 5.21.4-8caf727 from Canonical** installed
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ lxd --version
5.21.4 LTS
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ sudo lxd init
Would you like to use LXD clustering? (yes/no) [default=no]:
Do you want to configure a new storage pool? (yes/no) [default=yes]:
Name of the new storage pool [default=default]:
Name of the storage backend to use (ceph, lvm, zfs, dir, powerflex, pure, btrfs) [default=zfs]: dir
Would you like to connect to a MAAS server? (yes/no) [default=no]:
Would you like to create a new local network bridge? (yes/no) [default=yes]:
What should the new bridge be called? [default=lxdbr0]:
What IPv4 address should be used? (CIDR subnet notation, "auto" or "none") [default=auto]:
What IPv6 address should be used? (CIDR subnet notation, "auto" or "none") [default=auto]:
Would you like the LXD server to be available over the network? (yes/no) [default=no]: ^C
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ sudo lxd init
Would you like to use LXD clustering? (yes/no) [default=no]:
Do you want to configure a new storage pool? (yes/no) [default=yes]:
Name of the new storage pool [default=default]:
Name of the storage backend to use (ceph, lvm, zfs, dir, powerflex, pure, btrfs) [default=zfs]: dir
Would you like to connect to a MAAS server? (yes/no) [default=no]:
Would you like to create a new local network bridge? (yes/no) [default=yes]:
What should the new bridge be called? [default=lxdbr0]:
What IPv4 address should be used? (CIDR subnet notation, "auto" or "none") [default=auto]:
What IPv6 address should be used? (CIDR subnet notation, "auto" or "none") [default=auto]: none
Would you like the LXD server to be available over the network? (yes/no) [default=no]:
Would you like stale cached images to be updated automatically? (yes/no) [default=yes]:
Would you like a YAML "lxd init" preseed to be printed? (yes/no) [default=no]:
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ newgrp lxd
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ sudo usermod -aG lxd $USER
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ newgrp lxd
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$
```




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```
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009: ~  
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ lxd --version  
5.21.4 LTS  
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ sudo lxd init  
Would you like to use LXD clustering? (yes/no) [default=no]:  
Do you want to configure a new storage pool? (yes/no) [default=yes]:  
Name of the new storage pool [default=default]:  
Name of the storage backend to use (ceph, lvm, zfs, dir, powerflex, pure, btrfs) [default=zfs]: dir  
Would you like to connect to a MAAS server? (yes/no) [default=no]:  
Would you like to create a new local network bridge? (yes/no) [default=yes]:  
What should the new bridge be called? [default=lxdbr0]:  
What IPv4 address should be used? (CIDR subnet notation, "auto" or "none") [default=auto]:  
What IPv6 address should be used? (CIDR subnet notation, "auto" or "none") [default=auto]:  
Would you like the LXD server to be available over the network? (yes/no) [default=no]: ^C  
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ sudo lxd init  
Would you like to use LXD clustering? (yes/no) [default=no]:  
Do you want to configure a new storage pool? (yes/no) [default=yes]:  
Name of the new storage pool [default=default]:  
Name of the storage backend to use (ceph, lvm, zfs, dir, powerflex, pure, btrfs) [default=zfs]: dir  
Would you like to connect to a MAAS server? (yes/no) [default=no]:  
Would you like to create a new local network bridge? (yes/no) [default=yes]:  
What should the new bridge be called? [default=lxdbr0]:  
What IPv4 address should be used? (CIDR subnet notation, "auto" or "none") [default=auto]:  
What IPv6 address should be used? (CIDR subnet notation, "auto" or "none") [default=auto]: none  
Would you like the LXD server to be available over the network? (yes/no) [default=no]:  
Would you like stale cached images to be updated automatically? (yes/no) [default=yes]:  
Would you like a YAML "lxd init" preseed to be printed? (yes/no) [default=no]:  
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ newgrp lxd  
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ sudo usermod -aG lxd $USER  
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ newgrp lxd  
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ lxc launch ubuntu:22.04 dev-container  
Launching dev-container  
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ lxc list  
+-----+-----+-----+-----+-----+-----+  
| NAME | STATE | IPV4 | IPV6 | TYPE | SNAPSHOTS |  
+-----+-----+-----+-----+-----+-----+  
| dev-container | RUNNING | 10.213.205.233 (eth0) | | CONTAINER | 0 |  
+-----+-----+-----+-----+-----+-----+  
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$
```

```
root@dev-container: ~  
Name of the new storage pool [default=default]:  
Name of the storage backend to use (ceph, lvm, zfs, dir, powerflex, pure, btrfs) [default=zfs]: dir  
Would you like to connect to a MAAS server? (yes/no) [default=no]:  
Would you like to create a new local network bridge? (yes/no) [default=yes]:  
What should the new bridge be called? [default=lxdbr0]:  
What IPv4 address should be used? (CIDR subnet notation, "auto" or "none") [default=auto]:  
What IPv6 address should be used? (CIDR subnet notation, "auto" or "none") [default=auto]: none  
Would you like the LXD server to be available over the network? (yes/no) [default=no]:  
Would you like stale cached images to be updated automatically? (yes/no) [default=yes]:  
Would you like a YAML "lxd init" preseed to be printed? (yes/no) [default=no]:  
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ newgrp lxd  
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ sudo usermod -aG lxd $USER  
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ newgrp lxd  
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ lxc launch ubuntu:22.04 dev-container  
Launching dev-container  
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ lxc list  
+-----+-----+-----+-----+-----+-----+  
| NAME | STATE | IPV4 | IPV6 | TYPE | SNAPSHOTS |  
+-----+-----+-----+-----+-----+-----+  
| dev-container | RUNNING | 10.213.205.233 (eth0) | | CONTAINER | 0 |  
+-----+-----+-----+-----+-----+-----+  
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ lxc exec dev-container -- bash  
root@dev-container:~# hostname  
dev-container  
root@dev-container:~# cat /etc/os-release  
PRETTY_NAME="Ubuntu 22.04.5 LTS"  
NAME="Ubuntu"  
VERSION_ID="22.04"  
VERSION="22.04.5 LTS (Jammy Jellyfish)"  
VERSION_CODENAME=jammy  
ID=ubuntu  
ID_LIKE=debian  
HOME_URL="https://www.ubuntu.com/"  
SUPPORT_URL="https://help.ubuntu.com/"  
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"  
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"  
UBUNTU_CODENAME=jammy  
root@dev-container:~#
```



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```
root@dev-container: ~  
Get:9 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [5238 kB]  
Get:10 http://security.ubuntu.com/ubuntu jammy-security/restricted Translation-en [998 kB]  
Get:11 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [1018 kB]  
Get:12 http://security.ubuntu.com/ubuntu jammy-security/universe Translation-en [225 kB]  
Get:13 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 c-n-f Metadata [22.7 kB]  
Get:14 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 Packages [51.9 kB]  
Get:15 http://security.ubuntu.com/ubuntu jammy-security/multiverse Translation-en [10.6 kB]  
Get:16 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 c-n-f Metadata [388 B]  
Get:17 http://archive.ubuntu.com/ubuntu jammy/universe Translation-en [5652 kB]  
Get:18 http://archive.ubuntu.com/ubuntu jammy/universe amd64 c-n-f Metadata [286 kB]  
Get:19 http://archive.ubuntu.com/ubuntu jammy/multiverse amd64 Packages [217 kB]  
Get:20 http://archive.ubuntu.com/ubuntu jammy/multiverse Translation-en [112 kB]  
Get:21 http://archive.ubuntu.com/ubuntu jammy/multiverse amd64 c-n-f Metadata [8372 B]  
Get:22 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3285 kB]  
Get:23 http://archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [499 kB]  
Get:24 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 c-n-f Metadata [19.2 kB]  
Get:25 http://archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [5413 kB]  
Get:26 http://archive.ubuntu.com/ubuntu jammy-updates/restricted Translation-en [1028 kB]  
Get:27 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1256 kB]  
Get:28 http://archive.ubuntu.com/ubuntu jammy-updates/universe Translation-en [314 kB]  
Get:29 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 c-n-f Metadata [30.4 kB]  
Get:30 http://archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 Packages [59.0 kB]  
Get:31 http://archive.ubuntu.com/ubuntu jammy-updates/multiverse Translation-en [13.5 kB]  
Get:32 http://archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 c-n-f Metadata [612 B]  
Get:33 http://archive.ubuntu.com/ubuntu jammy-backports/main amd64 Packages [69.4 kB]  
Get:34 http://archive.ubuntu.com/ubuntu jammy-backports/main Translation-en [11.5 kB]  
Get:35 http://archive.ubuntu.com/ubuntu jammy-backports/main amd64 c-n-f Metadata [412 B]  
Get:36 http://archive.ubuntu.com/ubuntu jammy-backports/restricted amd64 c-n-f Metadata [116 B]  
Get:37 http://archive.ubuntu.com/ubuntu jammy-backports/universe amd64 Packages [31.7 kB]  
Get:38 http://archive.ubuntu.com/ubuntu jammy-backports/universe Translation-en [16.9 kB]  
Get:39 http://archive.ubuntu.com/ubuntu jammy-backports/universe amd64 c-n-f Metadata [672 B]  
Get:40 http://archive.ubuntu.com/ubuntu jammy-backports/multiverse amd64 c-n-f Metadata [116 B]  
Fetched 43.8 MB in 6s (7754 kB/s)  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
4 packages can be upgraded. Run 'apt list --upgradable' to see them.  
root@dev-container:~#
```

```
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009: ~  
Get:40 http://archive.ubuntu.com/ubuntu jammy-backports/multiverse amd64 c-n-f Metadata [116 B]  
Fetched 43.8 MB in 6s (7754 kB/s)  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
4 packages can be upgraded. Run 'apt list --upgradable' to see them.  
root@dev-container:~# apt install git -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
Suggested packages:  
  git-daemon-run | git-daemon-sysvinit git-doc git-email git-gui gitk gitweb git-cvs git-mediawiki  
  git-svn  
The following packages will be upgraded:  
  git  
1 upgraded, 0 newly installed, 0 to remove and 3 not upgraded.  
Need to get 3174 kB of archives.  
After this operation, 4096 B disk space will be freed.  
Get:1 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 git amd64 1:2.34.1-1ubuntu1.17 [3174 kB]  
Fetched 3174 kB in 3s (1221 kB/s)  
(Reading database ... 34186 files and directories currently installed.)  
Preparing to unpack .../git_1%3a2.34.1-1ubuntu1.17_amd64.deb ...  
Unpacking git (1:2.34.1-1ubuntu1.17) over (1:2.34.1-1ubuntu1.16) ...  
Setting up git (1:2.34.1-1ubuntu1.17) ...  
Scanning processes...  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
root@dev-container:~# git --version  
git version 2.34.1  
root@dev-container:~# exit  
exit  
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$
```




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```
root@dev-container: ~  
Need to get 3174 kB of archives.  
After this operation, 4096 B disk space will be freed.  
Get:1 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 git amd64 1:2.34.1-1ubuntu1.17 [3174 kB]  
Fetched 3174 kB in 3s (1221 kB/s)  
(Reading database ... 34186 files and directories currently installed.)  
Preparing to unpack .../git_1%3a2.34.1-1ubuntu1.17_amd64.deb ...  
Unpacking git (1:2.34.1-1ubuntu1.17) over (1:2.34.1-1ubuntu1.16) ...  
Setting up git (1:2.34.1-1ubuntu1.17) ...  
Scanning processes...  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
root@dev-container:~# git --version  
git version 2.34.1  
root@dev-container:~# exit  
exit  
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ lxc list  
+-----+-----+-----+-----+-----+-----+  
| NAME | STATE | IPV4 | IPV6 | TYPE | SNAPSHOTS |  
+-----+-----+-----+-----+-----+-----+  
| dev-container | RUNNING | 10.213.205.233 (eth0) | | CONTAINER | 0 |  
+-----+-----+-----+-----+-----+-----+  
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ lxc stop dev-container  
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ lxc start dev-container  
tenperformer4@tenperformer4-Standard-PC-Q35-ICH9-2009:~$ lxc exec dev-container -- bash  
root@dev-container:~# apt update  
apt install docker.io -y  
Hit:1 http://security.ubuntu.com/ubuntu jammy-security InRelease  
Hit:2 http://archive.ubuntu.com/ubuntu jammy InRelease  
Hit:3 http://archive.ubuntu.com/ubuntu jammy-updates InRelease  
Hit:4 http://archive.ubuntu.com/ubuntu jammy-backports InRelease  
Reading package lists... Done  
Building dependency tree... Done
```

SonarQube

```
ubuntu@devops-instance: ~/LeetCode-Throne  
::]:3000->3000/tcp blitz-container  
ubuntu@devops-instance:~/LeetCode-Throne$ docker start blitz-container  
blitz-container  
ubuntu@devops-instance:~/LeetCode-Throne$ curl localhost:3000  
<!doctype html>  
<html lang="en">  
  <head>  
    <meta charset="UTF-8" />  
    <!-- <link rel="icon" type="image/svg+xml" href="/vite.svg" /> -->  
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />  
    <link rel="icon" type="image/png" href="https://tse3.mm.bing.net/th/id/OIP.1tecZz8ycQUqoyG091habgHaE  
8?pid=Api&P=0&h=180" />  
    <title>CodeThrone</title>  
    <script type="module" crossorigin src="/assets/index-CMnEYOJK.js"></script>  
    <link rel="stylesheet" crossorigin href="/assets/index-BGLfvPb5.css">  
  </head>  
  <body>  
    <div id="root"></div>  
  </body>  
</html>  
ubuntu@devops-instance:~/LeetCode-Throne$ docker run -d \  
--name sonarqube \  
-p 9000:9000 \  
sonarqube:lts-community  
Unable to find image 'sonarqube:lts-community' locally  
lts-community: Pulling from library/sonarqube  
60d98d907669: Pull complete  
e24a8b9e052f: Pull complete  
f3929ce9ef98: Pull complete  
1df735f481ad: Pull complete  
5d5a1fad7028: Pull complete  
eb27e3a98da1: Pull complete  
c7ed1fe61e07: Pull complete  
4f4f6700ef54: Pull complete  
Digest: sha256:f709975ab31d2d08f5a3ae2dc73a31ee011afc8cf28845082c17c55d45df9df5  
Status: Downloaded newer image for sonarqube:lts-community  
07ea7007e07ecc2675bdd29ee3f0baa9ee3375d95a54a3b761f08de8a06c18  
ubuntu@devops-instance:~/LeetCode-Throne$
```



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```
Activities Terminal Apr 6 10:46
ubuntu@devops-instance: ~/LeetCode-Throne

/sonarqube/temp -XX:-OmitStackTraceInFastThrow --add-opens=java.base/java.util=ALL-UNNAMED --add-exports=java.base/jdk.internal.ref=ALL-UNNAMED --add-opens=java.base/java.lang=ALL-UNNAMED --add-opens=java.base/java.nio=ALL-UNNAMED --add-opens=java.base/sun.nio.ch=ALL-UNNAMED --add-opens=java.management/sun.management=ALL-UNNAMED --add-opens=jdk.management/com.sun.management.internal=ALL-UNNAMED -Dcom.redhat.fips=false -Xmx512m -Xms128m -XX:+HeapDumpOnOutOfMemoryError -Dhttp.nonProxyHosts=localhost[127.*][:1] -cp ./lib/sonar-application-9.9.8.100196.jar:/opt/sonarqube/lib/jdbc/h2-2.1.214.jar org.sonar.ce.app.CeServer /opt/sonarqube/temp/sq-process2495785909685639080properties
2026.04.06 05:16:35 INFO web[[o.s.s.q.ProjectsInWarningDaemon] Counting number of projects in warning will be disabled as there are no more projects in warning.
2026.04.06 05:16:35 WARN app[[startup] #####
2026.04.06 05:16:35 WARN app[[startup] Default Administrator credentials are still being used. Make sure to change the password or deactivate the account.
2026.04.06 05:16:35 WARN app[[startup] #####
2026.04.06 05:16:35 INFO web[[o.s.s.p.Platform] Web Server is operational
WARNING: A terminally deprecated method in java.lang.System has been called
WARNING: System::setSecurityManager has been called by org.sonar.process.PluginSecurityManager (file:/opt/sonarqube/lib/sonar-application-9.9.8.100196.jar)
WARNING: Please consider reporting this to the maintainers of org.sonar.process.PluginSecurityManager
WARNING: System::setSecurityManager will be removed in a future release
2026.04.06 05:16:36 INFO ce[[o.s.p.ProcessEntryPoint] Starting Compute Engine
2026.04.06 05:16:36 INFO ce[[o.s.ce.app.CeServer] Compute Engine starting up...
2026.04.06 05:16:36 INFO ce[[o.s.sonar.db.Database] Create JDBC data source for jdbc:h2:tcp://127.0.0.1:9092/sonar;NON_KEYWORDS=VALUE
2026.04.06 05:16:36 INFO ce[[c.z.h.HikariDataSource] HikariPool-1 - Starting...
2026.04.06 05:16:36 INFO ce[[c.z.h.p.HikariPool] HikariPool-1 - Added connection conn0: url=jdbc:h2:tcp://127.0.0.1:9092/sonar user=
2026.04.06 05:16:36 INFO ce[[c.z.h.HikariDataSource] HikariPool-1 - Start completed.
2026.04.06 05:16:36 WARN ce[[o.s.db.dialect.H2] H2 database should be used for evaluation purpose only
2026.04.06 05:16:38 INFO ce[[o.s.s.p.ServerFileSystemImpl] SonarQube home: /opt/sonarqube
2026.04.06 05:16:39 INFO ce[[o.s.c.c.CePluginRepository] Load plugins
2026.04.06 05:16:40 INFO ce[[o.s.c.c.ComputeEngineContainerImpl] Running Community edition
2026.04.06 05:16:40 INFO ce[[o.s.ce.app.CeServer] Compute Engine is started
2026.04.06 05:16:41 INFO app[[o.s.a.SchedulerImpl] Process[ce] is up
2026.04.06 05:16:41 INFO app[[o.s.a.SchedulerImpl] SonarQube is operational
```

Activities Firefox Web Browser Apr 6 10:47

SonarQube x +

10.20.155.17:9000/sessions/new?return_to=%2F

Firefox automatically sends some data to Mozilla so that we can improve your experience. Choose What I Share

Log in to SonarQube

admin

.....

Log In Cancel

SonarQube™ technology is powered by SonarSource SA
[LGPL v3](#) - [Community](#) - [Documentation](#) - [Plugins](#)



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The image displays two screenshots of the SonarQube web interface, accessed via a Firefox Web Browser on April 6 at 10:48. The browser address bar shows the URL `10.20.155.17:9000/projects/create`.

Top Screenshot: The page title is "How do you want to create your project?". It features a navigation bar with links: Projects, Issues, Rules, Quality Profiles, Quality Gates, and Administration. A search bar is present on the right. The main content area asks, "How do you want to create your project?" and provides instructions: "Do you want to benefit from all of SonarQube's features (like repository import and Pull Request decoration)? Create your project from your favorite DevOps platform. First, you need to set up a DevOps platform configuration." Below this, there are five buttons for different platforms: "From Azure DevOps", "From Bitbucket Server", "From Bitbucket Cloud", "From GitHub", and "From GitLab". Each button has a sub-label "Set up global configuration". At the bottom, there is a link "Are you just testing or have an advanced use-case? Create a project manually." and a sidebar with a "Get the most out of SonarQube!" section.

Bottom Screenshot: The page title is "How do you want to create your project?". The URL is `10.20.155.17:9000/projects/create?mode=manual`. The main content area shows the "Manual Project Creation" form. It includes the following fields and labels:

- Project display name ***: Input field with "CodeThrone" and a green checkmark. Below it, a note says "Up to 255 characters. Some scanners might override the value you provide."
- Project key ***: Input field with "codethrone" and a green checkmark. Below it, a note says "The project key is a unique identifier for your project. It may contain up to 400 characters. Allowed characters are alphanumeric, '-' (dash), '_' (underscore), '.' (period) and ':' (colon), with at least one non-digit."
- Main branch name ***: Input field with "main". Below it, a note says "The name of your project's default branch" with a link to "Learn More".

At the bottom of the form, there is a "Set Up" button. A sidebar on the right contains a "Get the most out of SonarQube!" section with a "Learn More" button and a "Dismiss" button.



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Firefox Web Browser
Apr 6 10:50
CodeThrone
10.20.155.17:9000/dashboard?id=codethrone&selectedTutorial=local
Firefox automatically sends some data to Mozilla so that we can improve your experience. Choose What I Share
You're running a version of SonarQube that is no longer active. Please upgrade to an active version immediately. Learn More
sonarqube Projects Issues Rules Quality Profiles Quality Gates Administration
CodeThrone main
Overview Issues Security Hotspots Measures Code Activity Project Settings
Analyze your project
We initialized your project on SonarQube, now it's up to you to launch analyses!
1 Provide a token
Generate a project token
Token name Expires in
Analyze "CodeThrone" 30 days Generate
Please note that this token will only allow you to analyze the current project. If you want to use the same token to analyze multiple projects, you need to generate a global token in your user account. See the documentation for more information.
The token is used to identify you when an analysis is performed. If it has been compromised, you can revoke it at any point in time in your user account.
2 Run analysis on your project
Get the most out of SonarQube!
Take advantage of the whole ecosystem by using SonarLint, a free IDE plugin that helps you find and fix issues earlier in your workflow. Connect SonarLint to SonarQube to sync rule sets and issue states. Learn More Dismiss

Token SonarQube : `sqp_d60bf56e30edf2aad23ef960bb08e448243a417a`

Firefox Web Browser
Apr 6 10:51
CodeThrone
10.20.155.17:9000/dashboard?id=codethrone&selectedTutorial=local
Firefox automatically sends some data to Mozilla so that we can improve your experience. Choose What I Share
You're running a version of SonarQube that is no longer active. Please upgrade to an active version immediately. Learn More
sonarqube Projects Issues Rules Quality Profiles Quality Gates Administration
CodeThrone main
Overview Issues Security Hotspots Measures Code Activity Project Settings
Analyze your project
We initialized your project on SonarQube, now it's up to you to launch analyses!
1 Provide a token
Analyze "CodeThrone": sqp_d60bf56e30edf2aad23ef960bb08e448243a417a
The token is used to identify you when an analysis is performed. If it has been compromised, you can revoke it at any point in time in your user account.
Continue
2 Run analysis on your project
Get the most out of SonarQube!
Take advantage of the whole ecosystem by using SonarLint, a free IDE plugin that helps you find and fix issues earlier in your workflow. Connect SonarLint to SonarQube to sync rule sets and issue states. Learn More Dismiss
Embedded database should be used for evaluation.
The embedded database will not scale, it will not support production engine.



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```
Activities Terminal Apr 6 10:51
ubuntu@devops-instance: ~/LeetCode-Throne

2026.04.06 05:18:04 INFO web[AZ1h0USNgp2PTvrsAAAE][o.s.u.c.UpdateCenter] The plugin 'xml' version : 2.7
.0.3820 has not been found on the update center.
^Cubuntu@devops-instance:~/LeetCode-Throne$ sudo apt install unzip -y
wget https://binaries.sonarsource.com/Distribution/sonar-scanner-cli/sonar-scanner-cli-5.0.1.3006-linux.
zip
unzip sonar-scanner-cli-*.zip
sudo mv sonar-scanner-* /opt/sonar-scanner
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  zip
The following NEW packages will be installed:
  unzip
0 upgraded, 1 newly installed, 0 to remove and 1 not upgraded.
Need to get 174 kB of archives.
After this operation, 384 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 unzip amd64 6.0-28ubuntu4.1 [174 kB]
Fetched 174 kB in 1s (139 kB/s)
Selecting previously unselected package unzip.
(Reading database ... 75200 files and directories currently installed.)
Preparing to unpack .../unzip_6.0-28ubuntu4.1_amd64.deb ...
Unpacking unzip (6.0-28ubuntu4.1) ...
Setting up unzip (6.0-28ubuntu4.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
```

```
Activities Terminal Apr 6 10:53
ubuntu@devops-instance: ~/LeetCode-Throne

DITIONAL_LICENSE_INFO
sonar-scanner-5.0.1.3006-linux/jre/legal/java.instrument/LICENSE -> ../java.base/LICENSE
sonar-scanner-5.0.1.3006-linux/jre/legal/java.instrument/ASSEMBLY_EXCEPTION -> ../java.base/ASSEMBLY_E
XCEPTION
sonar-scanner-5.0.1.3006-linux/jre/legal/java.instrument/ADDITIONAL_LICENSE_INFO -> ../java.base/ADDIT
IONAL_LICENSE_INFO
sonar-scanner-5.0.1.3006-linux/jre/legal/java.sql.rowset/LICENSE -> ../java.base/LICENSE
sonar-scanner-5.0.1.3006-linux/jre/legal/java.sql.rowset/ASSEMBLY_EXCEPTION -> ../java.base/ASSEMBLY_E
XCEPTION
sonar-scanner-5.0.1.3006-linux/jre/legal/java.sql.rowset/ADDITIONAL_LICENSE_INFO -> ../java.base/ADDIT
IONAL_LICENSE_INFO
sonar-scanner-5.0.1.3006-linux/jre/legal/jdk.security.auth/LICENSE -> ../java.base/LICENSE
sonar-scanner-5.0.1.3006-linux/jre/legal/jdk.security.auth/ASSEMBLY_EXCEPTION -> ../java.base/ASSEMBLY
_EXCEPTION
sonar-scanner-5.0.1.3006-linux/jre/legal/jdk.security.auth/ADDITIONAL_LICENSE_INFO -> ../java.base/ADD
ITIONAL_LICENSE_INFO
sonar-scanner-5.0.1.3006-linux/jre/legal/jdk.management.agent/LICENSE -> ../java.base/LICENSE
sonar-scanner-5.0.1.3006-linux/jre/legal/jdk.management.agent/ASSEMBLY_EXCEPTION -> ../java.base/ASSEM
BLY_EXCEPTION
sonar-scanner-5.0.1.3006-linux/jre/legal/jdk.management.agent/ADDITIONAL_LICENSE_INFO -> ../java.base/
ADDITIONAL_LICENSE_INFO
sonar-scanner-5.0.1.3006-linux/jre/legal/java.rmi/LICENSE -> ../java.base/LICENSE
sonar-scanner-5.0.1.3006-linux/jre/legal/java.rmi/ASSEMBLY_EXCEPTION -> ../java.base/ASSEMBLY_EXCEPTIO
N
sonar-scanner-5.0.1.3006-linux/jre/legal/java.rmi/ADDITIONAL_LICENSE_INFO -> ../java.base/ADDITIONAL_L
ICENSE_INFO
mv: target '/opt/sonar-scanner': No such file or directory
ubuntu@devops-instance:~/LeetCode-Throne$ sudo mkdir -p /opt/sonar-scanner
ubuntu@devops-instance:~/LeetCode-Throne$ ls
Dockerfile package-lock.json system-architecture.png
PROJECT_DOCUMENTATION.md package.json tailwind.config.js
README.md postcss.config.js test-image-upload.html
backend public tsconfig.app.json
deploy-package.json render.yaml tsconfig.json
eslint.config.js sonar-scanner-5.0.1.3006-linux tsconfig.node.json
high_level_design.pdf sonar-scanner-cli-5.0.1.3006-linux.zip vite.config.ts
index.html src
ubuntu@devops-instance:~/LeetCode-Throne$
```




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```
Activities Terminal Apr 6 10:53
ubuntu@devops-instance: ~/LeetCode-Throne

sonar-scanner-5.0.1.3006-linux/jre/legal/java.sql.rowset/ADDITIONAL_LICENSE_INFO -> ../java.base/ADDITIONAL_LICENSE_INFO
sonar-scanner-5.0.1.3006-linux/jre/legal/jdk.security.auth/LICENSE -> ../java.base/LICENSE
sonar-scanner-5.0.1.3006-linux/jre/legal/jdk.security.auth/ASSEMBLY_EXCEPTION -> ../java.base/ASSEMBLY_EXCEPTION
sonar-scanner-5.0.1.3006-linux/jre/legal/jdk.security.auth/ADDITIONAL_LICENSE_INFO -> ../java.base/ADDITIONAL_LICENSE_INFO
sonar-scanner-5.0.1.3006-linux/jre/legal/jdk.management.agent/LICENSE -> ../java.base/LICENSE
sonar-scanner-5.0.1.3006-linux/jre/legal/jdk.management.agent/ASSEMBLY_EXCEPTION -> ../java.base/ASSEMBLY_EXCEPTION
sonar-scanner-5.0.1.3006-linux/jre/legal/jdk.management.agent/ADDITIONAL_LICENSE_INFO -> ../java.base/ADDITIONAL_LICENSE_INFO
sonar-scanner-5.0.1.3006-linux/jre/legal/java.rmi/LICENSE -> ../java.base/LICENSE
sonar-scanner-5.0.1.3006-linux/jre/legal/java.rmi/ASSEMBLY_EXCEPTION -> ../java.base/ASSEMBLY_EXCEPTION
sonar-scanner-5.0.1.3006-linux/jre/legal/java.rmi/ADDITIONAL_LICENSE_INFO -> ../java.base/ADDITIONAL_LICENSE_INFO
mv: target '/opt/sonar-scanner': No such file or directory
ubuntu@devops-instance:~/LeetCode-Throne$ sudo mkdir -p /opt/sonar-scanner
ubuntu@devops-instance:~/LeetCode-Throne$ ls
Dockerfile      package-lock.json      system-architecture.png
PROJECT_DOCUMENTATION.md package.json             tailwind.config.js
README.md        postcss.config.js      test-image-upload.html
backend          public                  tsconfig.app.json
deploy-package.json render.yaml             tsconfig.json
eslint.config.js sonar-scanner-5.0.1.3006-linux tsconfig.node.json
high_level_design.pdf sonar-scanner-cli-5.0.1.3006-linux.zip vite.config.ts
index.html       src
ubuntu@devops-instance:~/LeetCode-Throne$ sudo mv sonar-scanner-5.0.1.3006-linux/* /opt/sonar-scanner/
ubuntu@devops-instance:~/LeetCode-Throne$ echo 'export PATH=$PATH:/opt/sonar-scanner/bin' >> ~/.bashrc
source ~/.bashrc
ubuntu@devops-instance:~/LeetCode-Throne$ sonar-scanner -v
INFO: Scanner configuration file: /opt/sonar-scanner/conf/sonar-scanner.properties
INFO: Project root configuration file: NONE
INFO: SonarScanner 5.0.1.3006
INFO: Java 17.0.7 Eclipse Adoptium (64-bit)
INFO: Linux 6.8.0-101-generic amd64
ubuntu@devops-instance:~/LeetCode-Throne$
```

```
Activities Terminal Apr 6 10:54
ubuntu@devops-instance: ~/LeetCode-Throne

GNU nano 7.2 sonar-project.properties *
sonar.projectKey=codethrone
sonar.projectName=CodeThrone
sonar.host.url=http://localhost:9000
sonar.login=YOUR_TOKEN_HERE
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo
^X Exit ^R Read File ^L Replace ^U Paste ^J Justify ^_ Go To Line M-E Redo



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```
Activities Terminal Apr 6 10:55
ubuntu@devops-instance: ~/LeetCode-Throne
GNU nano 7.2 sonar-project.properties
sonar.projectKey=codethrone
sonar.projectName=CodeThrone
sonar.host.url=http://localhost:9000
sonar.login=sqp_d60bf56e30edf2aad23ef960bb08e448243a417a
```

```
Activities Terminal Apr 6 10:56
ubuntu@devops-instance: ~/LeetCode-Throne
ubuntu@devops-instance:~/LeetCode-Throne$ sudo mkdir -p /opt/sonar-scanner
ubuntu@devops-instance:~/LeetCode-Throne$ ls
Dockerfile          package-lock.json    system-architecture.png
PROJECT_DOCUMENTATION.md package.json          tailwind.config.js
README.md           postcss.config.js   test-image-upload.html
backemd             public              tsconfig.app.json
deploy-package.json render.yaml          tsconfig.json
eslint.config.js    sonar-scanner-5.0.1.3006-linux tsconfig.node.json
high_level_design.pdf sonar-scanner-cli-5.0.1.3006-linux.zip vite.config.ts
index.html          src
ubuntu@devops-instance:~/LeetCode-Throne$ sudo mv sonar-scanner-5.0.1.3006-linux/* /opt/sonar-scanner/
ubuntu@devops-instance:~/LeetCode-Throne$ echo 'export PATH=$PATH:/opt/sonar-scanner/bin' >> ~/.bashrc
source ~/.bashrc
ubuntu@devops-instance:~/LeetCode-Throne$ sonar-scanner -v
INFO: Scanner configuration file: /opt/sonar-scanner/conf/sonar-scanner.properties
INFO: Project root configuration file: NONE
INFO: SonarScanner 5.0.1.3006
INFO: Java 17.0.7 Eclipse Adoptium (64-bit)
INFO: Linux 6.8.0-101-generic amd64
ubuntu@devops-instance:~/LeetCode-Throne$ nano sonar-project.properties
ubuntu@devops-instance:~/LeetCode-Throne$ sonar-scanner
INFO: Scanner configuration file: /opt/sonar-scanner/conf/sonar-scanner.properties
INFO: Project root configuration file: /home/ubuntu/LeetCode-Throne/sonar-project.properties
INFO: SonarScanner 5.0.1.3006
INFO: Java 17.0.7 Eclipse Adoptium (64-bit)
INFO: Linux 6.8.0-101-generic amd64
INFO: User cache: /home/ubuntu/.sonar/cache
INFO: Analyzing on SonarQube server 9.9.8.100196
INFO: Default locale: "en", source code encoding: "UTF-8" (analysis is platform dependent)
INFO: Load global settings
INFO: Load global settings (done) | time=152ms
INFO: Server id: 147B411E-AZ1h0DwACXs_TtPIu8o8I
INFO: User cache: /home/ubuntu/.sonar/cache
INFO: Load/download plugins
INFO: Load plugins index
INFO: Load plugins index (done) | time=95ms
INFO: Load/download plugins (done) | time=3352ms
```



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Activities Firefox Web Browser Apr 6 10:56

CodeThrone Projects

10.20.155.17:9000/projects

Firefox automatically sends some data to Mozilla so that we can improve your experience. Choose What I Share

You're running a version of SonarQube that is no longer active. Please upgrade to an active version immediately. Learn More

sonarqube Projects Issues Rules Quality Profiles Quality Gates Administration

Search for projects...

My Favorites All

Search by project name or key

1 project(s) Perspective: Overall Status Sort by: Name

CodeThrone

Project's Main Branch is not analyzed yet. Configure analysis

1 of 1 shown

Embedded database should be used for evaluation purposes only

Get the most out of SonarQube!

Take advantage of the whole ecosystem by using SonarLint, a free IDE plugin that helps you find and fix issues earlier in your workflow. Connect SonarLint to SonarQube to sync rule sets and issue states. Learn More Dismiss

Activities Firefox Web Browser Apr 6 10:57

CodeThrone Projects

10.20.155.17:9000/projects

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sonarqube Projects Issues Rules Quality Profiles Quality Gates Administration

Search for projects...

My Favorites All

Search by project name or key

1 project(s) Perspective: Overall Status Sort by: Name

CodeThrone Passed Last analysis: 1 minute ago

Bugs Vulnerabilities Hotspots Review Code Smells Coverage Duplications

A A E A — 0.0%

1 of 1 shown

Embedded database should be used for evaluation purposes only

The embedded database will not scale, it will not support upgrading to newer versions of SonarQube, migrating your data out of it into a different database engine.

SonarQube™ technology is powered by SonarSource SA

Community Edition - v9.9.8 (build 100196) NO LONGER ACTIVE - LGPL v3 - Community - Web API



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DevOps-Quality-Gate Rename Copy Set as Default Delete

This quality gate complies with Clean as You Code

This quality gate complies with the [Clean as You Code](#) methodology, so that you benefit from the most efficient approach to delivering Clean Code. It ensures that:

- No new bugs are introduced
- No new vulnerabilities are introduced
- All new security hotspots are reviewed
- New code has limited technical debt
- New code has limited duplication
- New code is properly covered by tests

Conditions ⓘ

Conditions on New Code

Metric	Operator	Value
Coverage	is less than	30.0%
Duplicated Lines (%)	is greater than	0.0%
Maintainability Rating	is worse than	A (Technical debt ratio is less than 5.0%)
Reliability Rating	is worse than	A (No bugs)
Security Hotspots Reviewed	is less than	100%
Security Rating	is worse than	A (No vulnerabilities)

You may click unlock to edit this quality gate. Adding extra conditions to a compliant quality gate can result in drawbacks. Are you reconsidering [Clean as You Code](#)? We strongly recommend this methodology to achieve a Clean Code status.

[Unlock editing](#)

Activities Firefox Web Browser Apr 6 10:58

CodeThrone x CodeThrone +

10.20.155.17:9000/dashboard?id=codethrone

Firefox automatically sends some data to Mozilla so that we can improve your experience. [Choose What I Share](#)

You're running a version of SonarQube that is no longer active. Please upgrade to an active version immediately. [Learn More](#)

sonarqube Projects Issues Rules Quality Profiles Quality Gates Administration Search for project

CodeThrone main Last analysis of this Branch had 2 warnings April 6, 2026 at 10:56

Overview Issues Security Hotspots Measures Code Activity Project Settings

To benefit from more of SonarQube's features, set up analysis in your favorite CI.

QUALITY GATE STATUS ⓘ

Passed

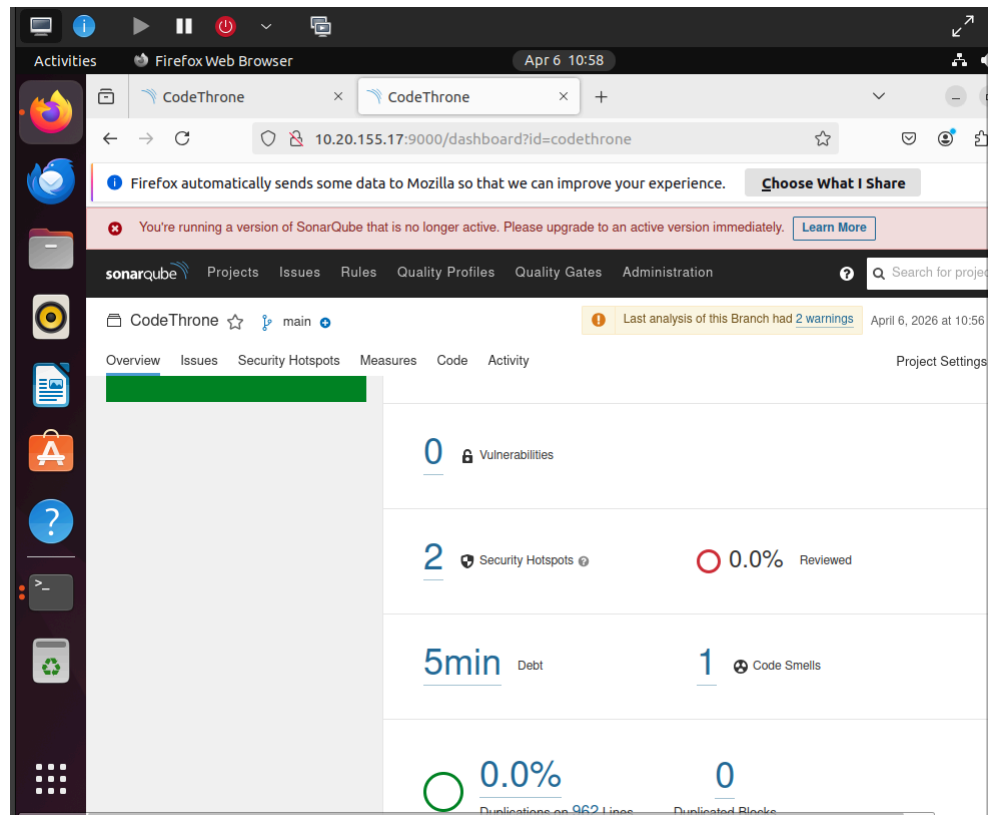
All conditions passed.

MEASURES

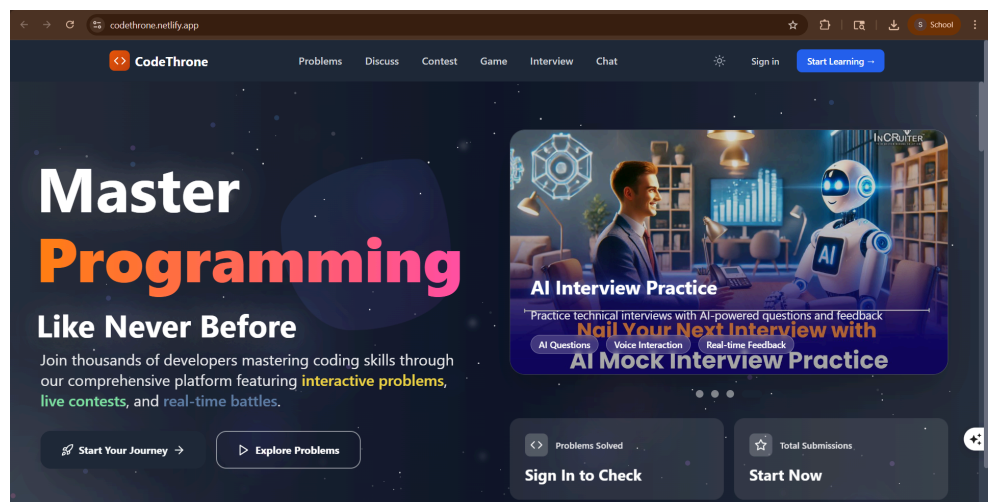
New Code	Overall Code
0 Bugs	
0 Vulnerabilities	
2 Security Hotspots ⓘ	0.0% Reviewed



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Final Project Output





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The screenshot shows the CodeThrone website's 'Problem of the Day' section. It features a star icon and the text 'Problem of the Day' with a subtext 'Solve today's challenge and earn coins!'. Below this, it says '+10 Coins' and 'Solve Now'. The problem title is 'Find First and Last Position of Element in Sorted Array' with tags 'Medium', 'Array', and 'Binary Search'. It also indicates '0 solved today'. A search bar and filter options are visible below the problem card.

This screenshot displays the detailed view of the 'Find First and Last Position of Element in Sorted Array' problem. It includes the problem description, examples, and a code editor. The code is written in C++ and implements a binary search algorithm to find the first and last positions of a target element in a sorted array.

```
1 #include <bits/stdc++.h>
2 using namespace std;
3
4 int findfirst(vector<int>& arr, int target) {
5     // Implement logic here
6 }
7
8 int findlast(vector<int>& arr, int target) {
9     // Implement logic here
10 }
11
12 int main() {
13     int n; cin >> n;
14     vector<int> arr(n);
15     for (int i = 0; i < n; i++) cin >> arr[i];
16     int target; cin >> target;
17     cout << findfirst(arr, target) << " " << findlast(arr, target);
18     return 0;
19 }
20
21
```

The screenshot shows the 'Programming Contests' section of the CodeThrone website. It features a trophy icon and the text 'Programming Contests' with a subtext 'Participate in exciting programming competitions and test your skills against other developers'. Below this, there are buttons for 'All Contests', 'Upcoming', 'Live', and 'Past'. A featured contest titled 'CODING COMPETITION' is highlighted, with details about when, who, and where to participate.



ABOUT PROJECT:

Project Overview

CodeThrone is a full-stack web application designed to provide a comprehensive platform for competitive programming, learning, and collaboration. It integrates problem-solving, contests, real-time communication, and performance tracking into a unified system. The platform is developed using modern web technologies and follows a scalable and modular architecture approach to ensure efficiency and maintainability. The primary objective of CodeThrone is to enhance coding skills by providing an engaging and interactive environment where users can practice problems, participate in contests, and collaborate with others.

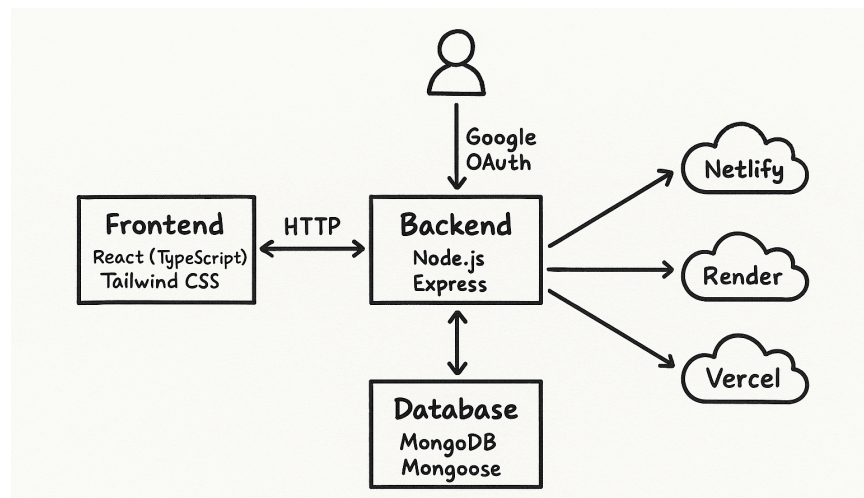
Key Features

The application offers a wide range of features, including:

- Coding contests with real-time leaderboards
- Problem-solving playground with integrated code editor
- Discussion forums and real-time chat system
- User profiles with performance tracking
- Admin dashboard for managing platform activities
- AI-powered interview preparation and assistance

These features collectively create a complete ecosystem for coding practice and competitive programming.

System Architecture





CodeThrone follows a modern, scalable architecture consisting of multiple layers:

1. Frontend Layer

The frontend is developed using React and TypeScript with Tailwind CSS. It provides an interactive and responsive user interface, ensuring seamless navigation and usability across devices.

2. Backend Layer

The backend is implemented using Node.js and Express, which handles business logic, API processing, and communication between frontend and database.

3. Database Layer

MongoDB is used as a NoSQL database to store user data, problems, contests, and chat information. Mongoose is used for schema management and data validation.

4. Real-time Communication

Socket.io is integrated for real-time chat and live updates such as leaderboards and contest interactions.

5. Authentication System

Firebase and OAuth-based authentication are used to provide secure login functionality with JWT-based session management.

Working of the System

The working of CodeThrone can be described in the following flow:

1. User accesses the platform through a web browser
2. Frontend sends API requests to backend services
3. Backend processes the request and interacts with MongoDB
4. Data is returned to frontend and displayed dynamically
5. Real-time features such as chat and contests are handled using



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	WebSockets 6. Code execution is performed using external Judge0 API The system ensures smooth interaction between all components while maintaining performance and scalability. DevOps Integration in Project The CodeThrone project is enhanced using DevOps practices to automate development, testing, and deployment processes. ● Virtualization using KVM provides an isolated environment ● Docker is used for containerizing the application ● GitHub is used for version control and collaboration ● Jenkins automates the CI/CD pipeline ● SonarQube ensures code quality and security ● Multipass and LXD manage lightweight virtual machines and containers This integration ensures continuous delivery, efficient resource usage, and system reliability. Project Workflow (Integrated View) 1. Developer writes code and pushes to GitHub 2. Jenkins detects changes and triggers pipeline 3. Application is built and tested automatically 4. Docker container is created and deployed 5. SonarQube performs code quality analysis 6. Application runs inside VM/container environment This automated workflow ensures fast and reliable software delivery.
CONCLUSION:	The CodeThrone project successfully demonstrates the development of a scalable and feature-rich web application integrated with modern DevOps practices. The implementation combines frontend, backend, database, and real-time technologies to create a comprehensive competitive programming platform. The use of modular architecture and modern frameworks ensures maintainability, performance, and user-friendly interaction.



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	The integration of DevOps tools such as Docker, Jenkins, SonarQube, Multipass, and LXD enhances the overall efficiency of the system by enabling automation, continuous integration, and quality assurance. The project effectively showcases how modern development practices can be applied to build reliable, scalable, and production-ready applications while maintaining code quality and deployment efficiency.
--	---